Reviewed Preprint
v1 • August 26, 2025
Not revised

Reviewed Preprint
v2 • March 6, 2026
Revised by authors

For correspondence:
sunyh@ihb.ac.cn

Competing interests: No competing interests declared

Funding: See page 23

Reviewing editor: Erika A Bach, NYU Grossman School of Medicine, United States

Trpv4 links environmental temperature to testicular differentiation in hermaphroditic ricefield eel

Yang Zhi[1,2,3], **Luo Tingting**[1,2,3], **Zhang Yimin**[1,2,3], **Sun Yuhua**[1,2,3]

[1]Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China • [2]College of Advanced Agricultural Sciences, University of Chinese Academy of Sciences, Beijing, China • [3]Hubei Hongshan Lab, Wuhan, China

eLife Assessment

This study presents **useful** findings on the molecular mechanisms driving female-to-male sex reversal in the ricefield eel (Monopterus albus) during aging, which would be of interest to biologists studying sex determination. The manuscript describes an interesting mechanism potentially underlying sex differentiation in M. albus. However, the current data are **incomplete** and would benefit from more rigorous experimental approaches.

https://doi.org/10.7554/eLife.108272.2.sa2

Abstract

The ricefield eel (*Monopterus albus*), an economically important aquaculture species in China, is a freshwater teleost fish that exhibits protogynous hermaphroditism. Although progress has been made in understanding the sex determination and differentiation of this species, the underlying mechanisms remain unclear. Here we show that warm temperature promotes gonadal transformation by up-regulating testicular differentiation genes such as *dmrt1/sox9a* in ovaries. Trpv4, a Ca^{2+}-permeable cation channel expressed in gonadal somatic cells, is highly sensitive to ambient temperature and mediates warm temperature-driven sex change of ricefield eel. In female fish reared at cool temperature, injection of Trpv4 agonist into the ovaries leads to a significant up-regulation of testicular differentiation genes, and in female fish exposed to warm temperature, Trpv4 inhibition or *trpv4* siRNA knockdown suppresses warm temperature-induced male gene expression. pStat3 signaling is downstream of Trpv4 and transduces Trpv4-controlled calcium signaling into the sex determination cascades. Inhibition of pStat3 activity prevents the up-regulation of testicular differentiation genes by warm temperature treatment and ovarian injection of Trpv4 agonist, whereas activation of pStat3 is sufficient to induce the expression of male genes, in the presence of Trpv4 antagonist. pStat3 binds and activates *jmjd3/kdm6b*, an activator of the male gene *dmrt1*. Consistently, ovarian injection of Kdm6b inhibitor blocks the up-regulation of testicular differentiation genes by warm temperature exposure. We propose that environmental factors, such as temperature, promote gonadal transformation of ricefield eel by inducing the expression of male pathway genes in ovaries via the Trpv4-pStat3-Kdm6b-Dmrt1 axis. Our results provide new insights into the molecular mechanism underlying natural sex change of ricefield eel, which will be useful for sex control in aquaculture.

Highlights

- Warm temperature promotes gonadal transformation of ricefield eel
- Trpv4 links environmental temperature and the sex determination pathway
- pStat3 is downstream of Trpv4-controlled calcium signaling
- pStat3 binds and activates *kdm6b/jmjd3*

Introduction

Sex determination in animals is intriguing and fascinating. In mammals, sex is determined genetically (genotypic sex determination, GSD). In lower vertebrates such as fish and reptiles, however, sex regulators are diverse. Their sex can be influenced by various environmental factors, including temperature, pH, the breeding density and social status (Gutzke and Crews. 1988; Honeycutt et al. 2019; Mei and Gui, 2015; Todd et al., 2019), the so-called environmental sex determination (ESD). The temperature-dependent sex determination (TSD) is one of the best studied forms of ESD. In red-eared slider turtle (*Trachemys scripta*), American alligator (*Alligator mississippiensis*) and Atlantic silverside (*Menidia menidia*), sex is determined solely by the temperature during the thermosensitive period of embryogenesis. In Australian central bearded dragon (*Pogona Vitticeps*) and Nile tilapia (*Oreochromis niloticus*), which display GSD, temperature can override the genetic materials to control the gonadal sex differentiation (Deveson et al., 2017; Holleley et al., 2015). Analysis of expression data during embryogenesis of normal ZW females and temperature sex reversed ZZ females has provided important insights into temperature-driven sex determination in the bearded dragon (Whiteley et al., 2021). Irrespective of TSD or GSD+TE (temperature effects), the downstream components are fairly conserved, including the epigenetic factors such as *jmjd3*/*kdm6b* and sex determination genes such as *dmrt1* (Castelli et al., 2020; Lu et al., 2025; Martinez-Pacheco et al., 2024; Weber et al., 2020; Whiteley et al., 2020; Wu et al., 2024).

Most vertebrates, including the TSD reptiles, exhibit gonochorism. However, approximately 6% of fish species exhibit hermaphroditism, including protandrous, protogynous, and bidirectional hermaphroditism (Todd et al., 2016). The majority of them are marine fish, appearing in 27 families (Muncaster et al., 2013; Peng et al., 2020; Shao et al., 2014; Todd et al., 2019). Compared to marine fish, natural sex change in freshwater fish is very rare. The ricefield eel (*Monopterus albus*), also called Asian swamp eel, which belongs to the family Synbranchidae in the Synbranchiformes, was firstly discovered as a protogynous hermaphroditic fish by Liu (Liu, 1944; Luo et al., 2026; Zhou and Gui, 2016). The species begins life as a female and then develops into a male through an intersex stage, thus displaying a female-to-male sex reversal during aging. Females are small in size (< 25 cm), and during and after sex change, there is a gradual increase in body size (> 55 cm for the majority of males). Among the described teleost fish species, ricefield eel has the fewest chromosome pairs (n= 12) with the fewest number of chromosome arms (Cheng and Zhou, 2022), and is emerging as an important model animal for studying sex determination and differentiation as well as adaptive evolution (Ji et al., 2001). As early as the Ming Dynasty in ancient China in 1578, pharmacist Shi-Zhen Li has described the medicinal value of ricefield eel in treatment of human diseases in his famous pharmacy monograph, the Bencao Gangmu, also called “Compendium of Materia Medica” (Cheng and Zhou, 2022). Nowadays, ricefield eel has been developed as one of the most important economical fish in freshwater aquaculture in China, with annual production exceeding 300,000 tons (Song et al., 2022). Unfortunately, the population has declined rapidly in the wild due to degradation of natural environment and human activity such as overfishing. The reproductive mode of ricefield eel, which leads to much more females than males in spawning season, severely affects the sex ratio, and decreases the productivity of broodstock. Moreover, adult females lay limited eggs (~200) due to its small size, which is a limiting factor for massive production of seedling for aquaculture industry. Thus, the elucidation of the mechanisms underlying the sex change/sex determination is urgent, which will aid in developing strategies/techniques for sex control that would break the bottleneck in aquaculture industry (Wu et al., 2019).

The life history of ricefield eel implies that environmental factors initiate and promote the gonadal transformation via epigenetic mechanisms. Consistently, histone demethyltransfearse/methyltransfearse genes such as *kdm6b/kmt2* and DNA methylation enzyme genes such as *dnmt1/3* were dynamically expressed throughout the sex change process, and the expression levels of the master sex determination/differentiation genes are closely correlated to the levels of DNA and histone methylation, which can be impacted by environmental exposure (Fan et al., 2021, 2022; Jiang et al., 2021; 2022; Hu et al., 2022; Wang et al., 2020).

However, these epigenetic regulators are not inherently responsive to the environmental cues, implying that certain molecular sensors exist and serve as the link between environmental stimuli and the sex determination pathway. We have recently proposed that there is a temperature-induced sex reversal (TISR) mechanism in ricefield eel (Zhang et al., 2025), similar to that of embryonic bearded dragon (*P. Vitticeps*) (Whiteley et al., 2018). Importantly, isolated ovarian explants are responsive to temperature stimuli, suggesting that the perception of temperature is executed by certain sensors expressed in ovarian cells.

While preliminary data have suggested that the Ca^{2+}-permeable, non-selective cation channel Trpv4 (Transient Receptor Potential Vanilloid 4 channel) might be a potential thermosensor, how Trpv4-regulated signals are transduced into the sex determination cascades remains unclear (Zhang et al., 2025). Recent studies have shown that signal transducer and activator of transcription 3 (STAT3) is phosphorylated by warm temperature-evoked Ca^{2+} influx, and then binds and transcriptionally regulates *kdm6b*, thereby initiating the female pathway by repressing the male sex determination gene *dmrt1* (Weber et al., 2020). In this work, we hypothesized that ricefield eel Trpv4 may bridge environmental temperature and the sex differentiation pathway. By using small molecule agonist and antagonist of Trpv4 as well as siRNA-mediated knockdown of *trpv4*, we provided solid evidences that temperature-evoked Trpv4 activity promotes the sex change of ricefield eel via the downstream Ca^{2+}-pStat3-Kdm6b axis.

Results

Warm temperature promotes gonadal transformation

Natural populations of ricefield eel are mainly distributed across East and Southeast Asia. Previous studies have reported that at the onset of sex change, wild fish from different geographic populations and habitats vary in age, body weight and length. For instance, it is around 16 cm long (18-month-old) in Bandung area of Indonesia (Liem, 1963), and ~20 cm in Hainan and Guangzhou areas of southern China (Chan and Philips, 1967; Wang and Zeng, 2006), ~30 cm (2-year-old) in Wuhan area of central China (Wang et al., 2008), and 35-40 cm (3-year-old) in Tianjin area of northern China (Fan et al., 2017; 2021; Liu and Wang, 1987). The average annual temperature in Hainan, Guangzhou, Wuhan and Tianjin areas is approximately 25, 22, 17, and 13 °C, respectively (Figure 1A). This observation implied that higher temperature facilitates the sex change of ricefield eel. To directly investigate this, during June-July, 2024, we have obtained ~200 2-year-old wild ricefield eels from the southernmost Hainan and central Wuhan, and examined their gonads. We found that less than 5% of the fish from Wuhan area were intersex animals, whereas approximately 24% of the fish from Hainan area were in intersex stage (Figure 1B).

In Wuhan area (Hubei province, China), the reproductive season of wild ricefield eel usually runs from May to August, when the average monthly temperature is 25-33 °C (Figure S1A). Immediately after spawning, the females (2-year-old) may undergo extensive ovarian tissue degeneration and physiological change (Liu and Gu, 1950), which leads to an irreversible commitment to becoming male via an intersex stage (Figures S1B-C). This observation again supported that the onset of sex change of ricefield eels is closely related to the external environment, in particular the warm temperature.

The above observations prompted us to hypothesize that warm temperature plays an important role in driving the sex change of ricefield eel. To directly test this, long term temperature experiments were performed using females from Wuhan Area (Figure 1C). One-and-a-half-year-old females (about 50 g) were randomly divided into two groups, and reared at 25/26 °C (cool temperatures, CT) and 33/34 °C (warm temperatures, WT), for a period of 6 months. At day 30, 90, and 180, the gonadal sex of randomly selected fish from different group was determined by H&E staining and expression analysis of sex-biased genes (Figure 1D). The average body length and weight of ricefield eels were comparable between the WT group and the CT group (Figures S1D-E). In CT group at day 90, ~90% gonads were ovaries, and ~10% were ovotestes. In WT group, however, ~65% gonads were ovaries, and ~35% were ovotestes (Figure 1E). In CT group at day

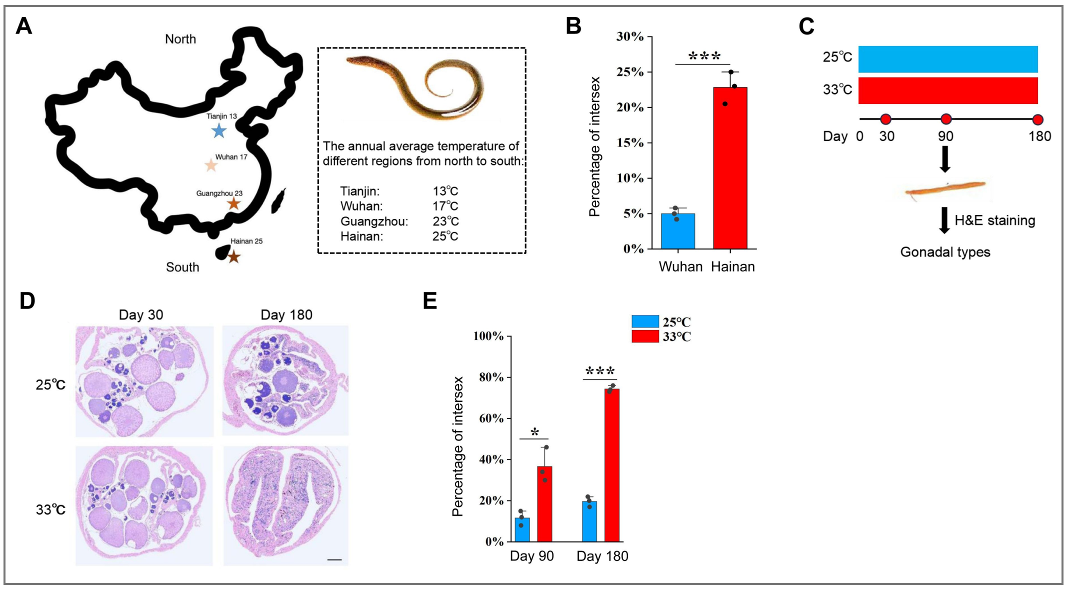


Figure 1. **Warm temperature promotes gonadal transformation in ricefield eel.**

(A) The distribution of 4 geographic populations of ricefield eels in China, showing the average annual temperature of each. (B) Bar graph showing the percentage of intersex animals in 2-year-old wild-caught ricefield eels from Hainan (Hainan province) and Wuhan area (Hubei Province). n=200 for each group. (C) Diagram showing the design of long term temperature experiments. Two temperatures were used: 25 °C (cool temperature), 33 °C (warm temperature). n=400 for each group. (D) Representative H&E staining images showing the gonad types of animals that were reared at cool and warm temperatures at the indicated time points. Bar: 200 μm. (E) Bar graph showing the percentage of intersex animals after 180 days of cool and warm temperature treatment. The experiments were repeated at least two times.

180, ~80% gonads were ovaries, and ~20% were ovotestes. In WT group, however, ~25% gonads were ovaries, and ~75% were ovotestes (Figure 1E). We concluded that warm temperature promotes gonadal transformation of ricefield eel.

Trpv4 is highly responding to environmental temperature

We went on to investigate how the gonadal tissues are responding to temperature cues. Previous work has suggested that Trpv4 associates environmental temperature and sex determination in TSD alligator and ricefield eel (Huang et al., 2024; Yatsu et al., 2015; Zhang et al., 2025). We hypothesized that ricefield eel Trpv4 is expressed in ovary and functions as a thermosensor that perceives/detects the environmental temperature cues. The results of qPCR experiments showed that *trpv4* was higher expressed in gonadal tissues than in non-gonadal tissues, exhibiting the highest expression in testis (Figure 2A). RNA in situ hybridization (ISH) experiments confirmed that *trpv4* levels increased from ovary to testis (Figure 2B), implying that it was functionally associated with testicular development.

In cultured primary ovarian explants, *trpv4* was one of the most up-regulated *trp* genes induced by warm temperature (Figure 2C). Temperature-shifting experiments showed that *trpv4* was highly sensitive to temperature cues (Figure 2D), displaying a pattern similar to the testicular differentiation genes such as *dmrt1/sox9a*, but inverse to that of the ovarian differentiation genes such as *cyp19a1a/foxl2* (Figure S2A). *trpv4* was elevated as early as 4 hours post warm temperature treatment (Figure S2B), suggesting that it was an early temperature-response gene. The *trpv4* gene encodes a constitutively active Ca^{2+}-permeable cation channel that can be activated by warm temperature (Güler et al., 2002). Consistently, when Ca^{2+} indicator Cal-520 was added in the cells, a significant increase in the calcium signal was observed at 34 °C in comparison to 26 °C (Figure 2E), which suggested that Ca^{2+} signaling mediates the temperature-controlled Trpv4 activity, similar to the embryonic dragon (Whiteley et al., 2021). This was further supported by our RNA-seq data that *trpv4* and many genes involved in Ca^{2+} transport and sequestration were up-regulated in ovotestes compared to ovaries (Zhang et al., 2025).

While *trpv4* is highly responding to temperature changes in cultured ovarian cells, it is not known whether this was the case in vivo. To explore this, female fish were transiently reared at cool temperature (25 °C) for 1 day, and then shifted to warm temperature (34 °C) for 10 days. *trpv4* mRNA in ovaries was already elevated on day 1 after shifting to 34 °C (Figure S2C), and its levels progressively increased over time, exhibiting a pattern similar to that of testicular differentiation genes (Figure 2F).

To understand in more detail of the role of *trpv4*, we studied its expression pattern in ovaries by performing ISH (mRNA in situ hybridization) experiment. The ISH results showed that *trpv4* was moderately detected in ovarian somatic cells around the oocytes at cool temperature (Figure 2G). Warm temperature exposure led to an increase of *trpv4* gene expression in ovarian follicles of different developmental stages, and its levels were positively correlated with the duration of temperature exposure. To explore the identity of the Trpv4-expressing follicle cells, double immunofluorescence (IF) experiments were performed using antibodies against Trpv4 and Foxl2, a granulosa cell marker. The IF results showed that Trpv4 protein was expressed in a portion of Foxl2-positive granulosa cells (Figure 2H). The observation suggested that these *trpv4*-expressing somatic cells in ovary may play an important role in gonadal transformation in response to warm temperature.

Temperature-induced male gene expression depends on Trpv4

The co-expression of *trpv4* and testicular differentiation genes prompted us to ask whether the thermosensor Trpv4 in ovaries function to transduce the temperature cues into the sex determination cascades. To investigate this, animal experiments were performed by injecting into ovaries with small molecules RN1734 and GSK1016790A, known Trpv4 specific antagonist and agonist, respectively (Liu et al., 2021). The experimental females were reared and divided into 4

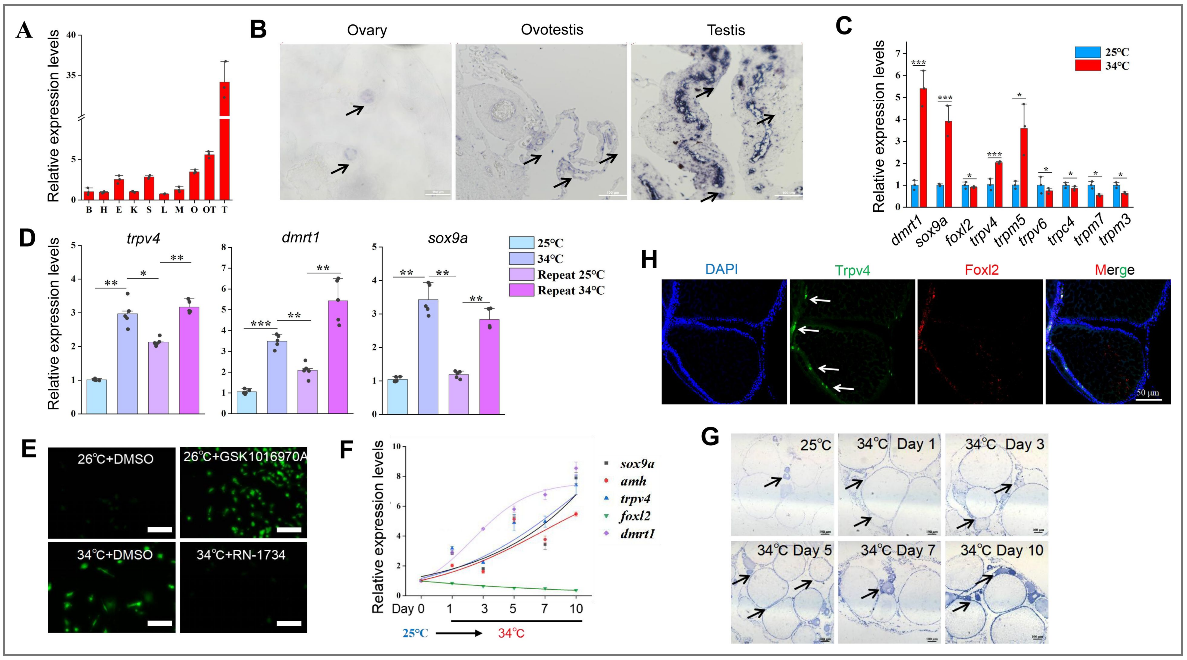


Figure 2. ***trpv4* is highly responding to environmental temperatures.**

(A) Relative expression levels of *trpv4* in 10 different tissues in adult ricefield eels. B: brain, H: heart, E: eye, K: kidney, S: spleen, L: liver, M:muscle, O: ovary, OT: ovotestis, T: testis. n=3. (B) ISH images showing *trpv4* expression in ovaries, ovotestes, and testes. Black arrows pointing to *trpv4* expressing cells. Bar: 200 μm. (C) The expression of the indicated *trp* and sex-biased genes of in vitro cultured ovaries at cool and warm temperatures. n=5 per group. (D) qPCR results showing the expression patterns of *trpv4* and male sex genes in repeated temperature shifting experiments of in vitro cultured ovaries. n=5 per group. (E) Confocal images showing the calcium signaling in cultured ovarian cells at the indicated conditions. After shifting to 34 °C for 2 h, Cal-520 was added and calcium signal imaging was performed. The Trpv4 agonist GSK1016970A and antagonist RN1734 were administrated in the culture medium, and 2 h later, calcium signal imaging was performed. Bar: 100 μm. (F) qPCR results showing the dynamic expression of the indicated genes in gonads of female ricefield eel at 25 °C, and at the indicated time points after shifting to 34 °C. day 1: day 1 after shifting to 34 °C. n=5 per group. (G) ISH images showing the dynamic expression of *trpv4* at the indicated time points before and after shifting to 34 °C. At 25 °C, *trpv4* was moderately expressed in follicles of various stages of developing oocytes, and interstitial cell types. After shifting to 34 °C, *trpv4* signals became much stronger compared to 25 °C. Bar: 200 μm. (H) Representative IF images showing the co-localization of Trpv4– and Foxl2-expressing cells. Bar: 50 μm. n=10. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. All experiments were repeated at least three times.

groups based on the combination of temperatures and the injected small molecules: 25 °C+ DMSO; 25 °C+ GSK1016790A; 34 °C+ DMSO; 34 °C+ RN1734 (Figure 3A). 3-10 days post injection, the gonads were isolated and used for the subsequent analyses.

The qPCR results showed that compared to the cool temperature group, warm temperature exposure increased the expression of testicular differentiation genes, accompanied by moderately decreased expression of ovarian differentiation genes (Figure 3B). Injection of 10 μM RN1734 blocked the up-regulation of testicular differentiation genes by warm temperature treatment, whereas injection of 0.1 μM GSK1016790A at 25 °C was sufficient to induce the expression of testicular differentiation genes to an extent similar to warm temperature treatment, at the expense of ovarian differentiation genes. The effects of small molecules on gene expression were dose dependent (Figure S3A). Similar results were observed at the protein levels, as revealed by Western blot (WB) and IF experiments (Figures 3C-F). We also repeated the experiments using cultured ovarian explants and/or cells, which produced similar results (Figures S3B-F). Moreover, activation and inhibition of Trpv4 ion channel function by the small molecules was correlated with increased and decreased calcium signaling, respectively (Figure 2E).

The above data suggested that Trpv4 is sufficient and necessary to induce the expression of testicular differentiation genes in ovaries, via its ion channel function. To further demonstrate this, *trpv4* siRNA was injected into the ovaries. *trpv4* expression was markedly down-regulated by 0.1 μM siRNA injection (Figure 4A). Importantly, siRNA injection led to a marked decrease in expression of testicular differentiation genes that were up-regulated by warm temperature exposure, and a slight increase in expression of ovarian differentiation genes that were repressed by warm temperature treatment. Similar results were observed at the protein levels as revealed by WB experiments (Figures 4B-C). We concluded that warm temperature-induced male gene expression is dependent on Trpv4 activity.

Next, we evaluated the effect of Trpv4 inhibition in testes of male adults. To this end, 10 μM RN1734 was injected into gonads of male fish reared at 25 °C, and qPCR was performed to assess the expression of female– and male pathway genes. The results showed that Trpv4 inhibition by RN1734 treatment led to an increase in the expression of ovarian differentiation genes in testes, at the expense of testicular differentiation genes (Figure 4D). The results suggested that Trpv4 activity is persistently required for the expression of testicular differentiation genes in male gonad.

pStat3 signaling is downstream of Trpv4

We next asked how Trpv4-controlled Ca^{2+} flux was interpreted and transduced into the sex determination cascades. In TSD turtles and TISR central bearded dragon, Stat3/4 are activated through phosphorylation by temperature-evoked calcium signaling, and is directly involved in sex determination (Weber et al., 2020; Whiteley et al., 2021). Based on our RNA-seq data, we found that s*tat3*, and the Jak/Stat3 pathway target genes such as *socs3* and *egr1/2*, were significantly up-regulated in early ovotestes compared to ovaries (Figure 5A). We therefore hypothesized that pStat3 signaling is downstream of Trpv4-contolled calcium influx to promote the expression of male pathway genes in ovaries. In fact, several lines of evidences were in favor of this hypothesis. First, activation of Trpv4 ion channel by GSK1016790A at cool temperature led to elevated pStat3 levels and calcium signals in ovarian explants, similar to that by higher temperature treatment, and inhibition of Trpv4 ion channel by RN1734 at warm temperature decreased pStat3 levels and calcium signals (Figures 5B-D). Second, pStat3 signals were detected in the gonadal somatic cells around the oocytes and interstitial cells, analogous to that of *trpv4* (Figure 5B; Figure 2G). Third, the levels of pStat3 were gradually increased from ovaries to testes in wild-caught ricefield eels, along with the male sex promoting factors such as Amh (Figures 5E-F; Figure S5A-B). Fourth, WB blot analysis of A23187– or BAPTA-AM-treated ovarian cells showed that exposure to the ionophore A23187 increased phosphorylation levels of Stat3 at 25 °C, whereas chelatin of calcium with BAPTA-AM led to diminished activation of Stat3 (Figures 5G-H). Collectively, these observations strongly suggested that Trpv4 cell autonomously controls phosphorylation of Stat3 via the regulation of calcium signal in gonadal somatic cells. To explore the function of the pStat3

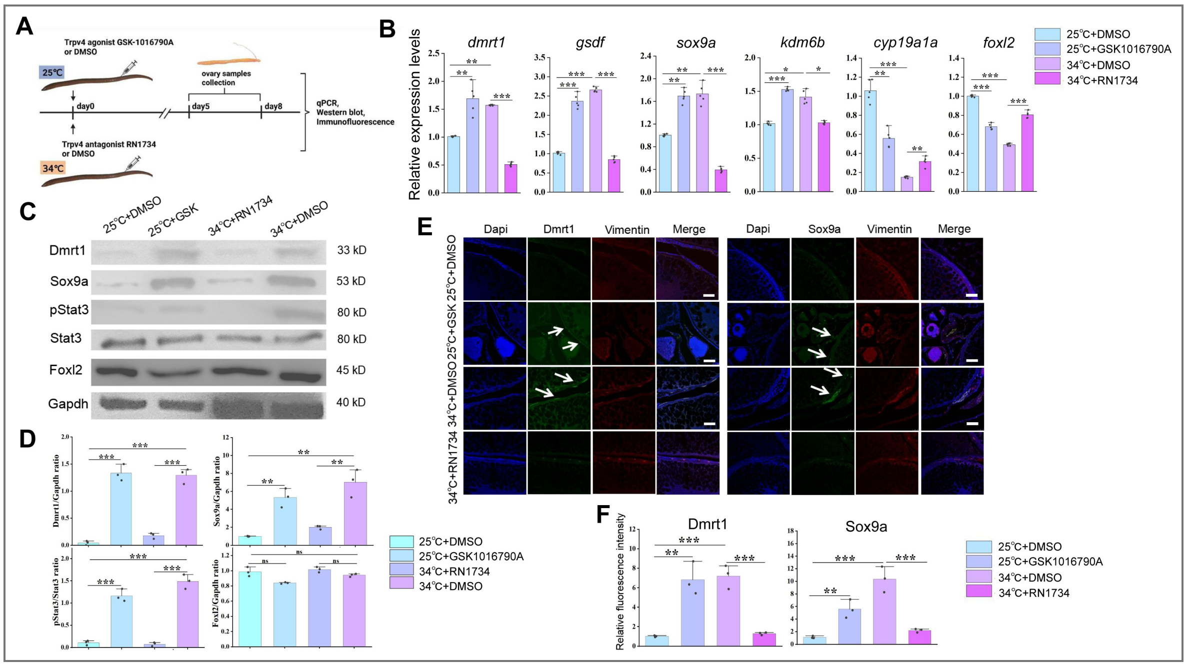


Figure 3. **Warm temperature-induced male gene expression depends on Trpv4.**

(A) Cartoon showing the design of animal experiments. Female eels kept at cool (25 °C) and warm (34 °C) temperatures were injected with the Trpv4 agonist GSK1016790A and antagonist RN1734 into the ovaries, respectively. After 2-3 days of injection, the ovaries were isolated and processed for the subsequent experiments. n=40. (B) qPCR results showing the relative expression levels of the sex-biased genes at the indicated conditions, based on the animal experiments. n=5 per group. (C) Representative WB images showing the expression of the indicated markers at the indicated conditions. GSK: GSK1016790A. (D) Relative quantification of the indicated proteins in Panel C. WB were repeated 3 times. (E) Representative IF images showing the expression of the indicated markers at the indicated conditions. Vimentin was used for show all cell types in ovaries. GSK: GSK1016790A. Bar: 200 μm. n=10, and 9/10 showed induced expression of Dmrt1/Sox9a. (F) quantification of panel E. *: $P< 0.05$, **: $P< 0.01$, ***: $P< 0.001$, and ****: $P< 0.0001$. ns: not significant. All experiments were repeated at least three times.

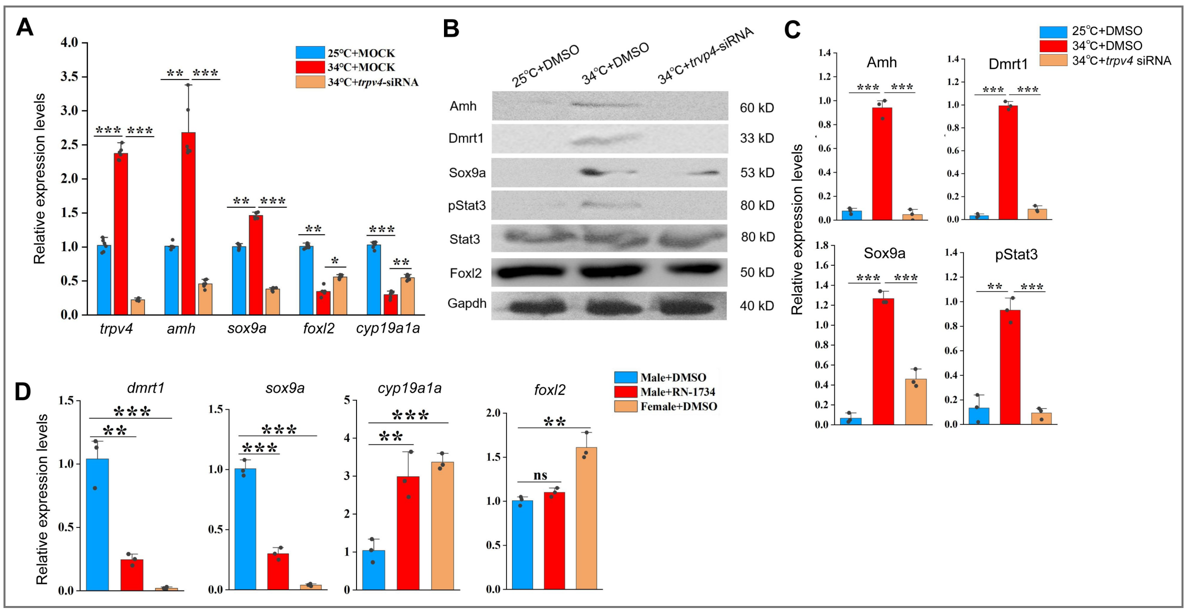


Figure 4. **siRNA-mediated *trpv4* knockdown abolishes the abnormal up-regulation of male genes by warm temperature treatment.**

(A) qPCR results showing the relative expression of the indicated genes at the indicated conditions in animal experiments. (B) Representative WB images showing the expression of sex biased proteins at the indicated conditions in animal experiments. (C) Quantification of panel B. (D) qPCR results showing the relative expression of the indicated genes at the indicated conditions. 10 μM RN1734 and DMSO was injected into gonads of male fish reared at 25/26 °C. Female fish injected with DMSO were used as control. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. All experiments were repeated at least three times.

signaling, animal experiments were performed by injecting into ovaries with small molecules HO-3867 or Colivelin. HO-3867, a curcumin analogue, is a selective pStat3 inhibitor, which blocks pStat3 activity by directly binding to Stat3 DNA binding domain, and Colivelin is a potent synthetic peptide activator of pStat3, which increases pStat3 levels by acting through the GP130/IL6ST complex (Wu et al., 2024). The injected females were divided into 4 groups based on the temperatures and the small molecules injected: 25 °C+ DMSO; 25 °C+ Colivelin; 34 °C+ DMSO; 34 °C+ HO-3867 (Figure 6A). 3-10 days post injection (dpj), the ovaries were isolated and subjected to the subsequent analyses. Injection of HO-3867 blocked the up-regulation of testicular differentiation genes by warm temperature exposure, whereas injection of Colivelin at 25 °C was sufficient to induce the expression of these genes to an extent similar to warm temperature treatment, at the expense of ovarian differentiation genes (Figures 6B). Similar results were observed at the protein levels as revealed by IF experiments (Figures 6C-D). We also repeated the experiments using ovarian explants and/or cells, which produced similar results (Figure S5A-B).

To functionally demonstrate that pStat3 signaling is downstream of Trpv4, rescue experiments were performed by injecting into ovaries with individual and combined small molecules. The females were divided into 6 groups based on the temperatures and the small molecules injected: 25 °C+ DMSO; 25 °C+ GSK1016790A; 25 °C+ GSK1016790A+HO-3867; 34 °C+ DMSO; 34 °C+ RN1734; 34 °C+ RN1734+ Colivelin. 3-5 days post injection, the ovaries were isolated and subjected to the qPCR analysis. The results showed that mRNA up-regulation of testicular differentiation genes by the administration of Trpv4 agonist at 25 °C was abolished by the treatment with pStat3 inhibitor, and that expression of testicular differentiation genes inhibited by Trpv4 antagonist treatment at 34 °C can be partially restored by the injection with pStat3 agonist (Figure 6E). Similar results were obtained using in vitro cultured ovarian explants (Figure S6C).

To further confirm that Trpv4 functions upstream of pStat3, Trpv4 agonist GSK1016790A and *stat3* siRNA were simultaneously injected into the ovaries of fish reared at 25 °C, and the expression of sex-biased genes was examined by qPCR analysis. The results showed that the testicular differentiating genes failed to be activated by GSK1016790A in the presence of *stat3* siRNAs (Figure 6F). Taken together, we concluded that Trpv4 promotes male sex gene expression in ovaries via the activation of the downstream pStat3 signaling.

pStat3 binds and activates kdm6b

Kdm6b is known a conserved activator at the top of the male sex determination pathway, via binding and activating *dmrt1* through removing the repressive histone mark H3K27me3 (Chen et al., 2024; Dupont et al., 2025; Ge et al, 2018; Lu et al., 2025; Yao et al;, 2024), and pStat3 can transcriptionally activate the *kdm6b* gene by directly binding to its upstream DNA motifs in reptiles (Wu et al., 2024; Weber et al., 2020). We reasoned that pStat3 had a similar function in ricefield eel. When analyzing –5 kb promoter sequences upstream the TSS of the *kdm6b* gene, we found that there were three conserved pStat3 binding sites (TTCnnnGAA) (Figure 7A). Chromatin immunoprecipitation (ChIP) experiments using pStat3 antibodies showed that pStat3 levels were significantly higher at *kdm6b* promoter in ovaries of females reared at warm temperature than at cool temperature, which was abolished in the presence of pStat3 inhibitor HO-3867 (Figure 7B). To explore whether pStat3 directly activates *kdm6b*, luciferase assay was performed. pGL4-*kdm6b*-luc construct was constructed by cloning ~3.5 kb *kdm6b* promoter sequences into the pGL4-luc plasmid. A mutant construct (pGL4-*kdm6b*M-luc) was generated by replacement of “TTCAGAGAA” with “TTAAAAGAA”. pGL4-*kdm6b*-luc and pGL4-*kdm6b*M-luc were transfected with into HEK293T cells in the presence and absence of pStat3 agonist Colivelin, and the luciferase activities were measured. The results showed that activities of pGL4-*kdm6b*-luc were significantly higher than that of pGL4-*kdm6b*M-luc in the presence of Colivelin (Figure 7C).

kdm6b mRNA was expressed in immature oocytes in ovaries, was induced by warm temperature exposure, and higher expressed in testes than in ovaries in ricefield eel (Figure S2C; Figure S3B; Figure S5A; Figure 6B; Figure 7D), analogous to that of *trpv4*. The expression of *kdm6b* was elevated by the injection (animal experiments) or addition (cell culture experiments)

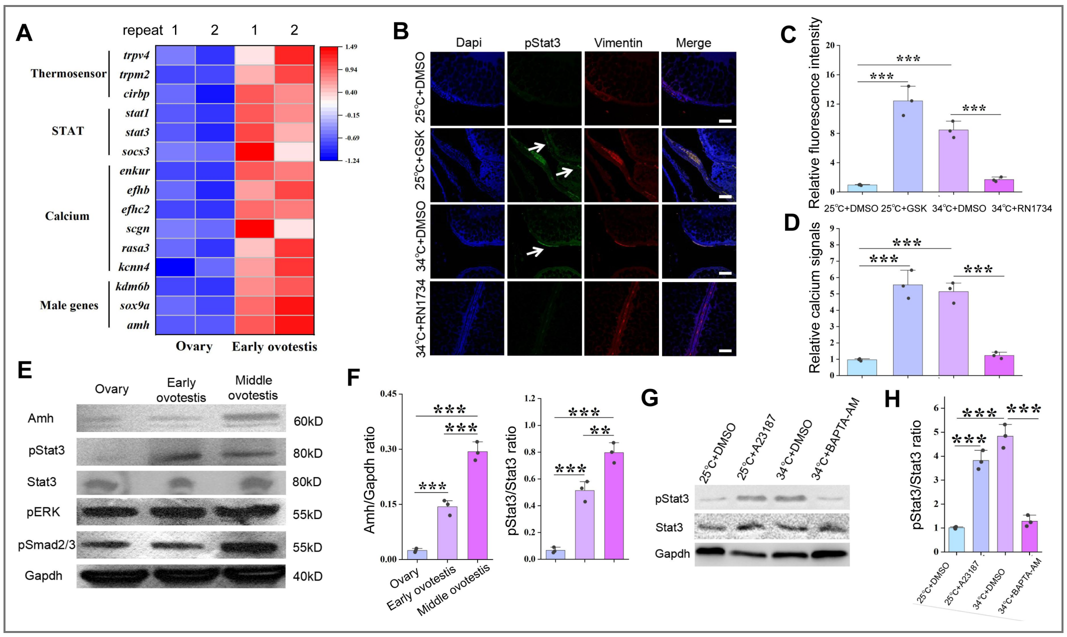


Figure 5. The JAK/Stat3 signaling is downstream of Trpv4.

(A) Heat map showing the expression of the indicated genes of different groups. (B) IF images showing the expression of pStat3 at the indicated conditions in animal experiments. The white arrows indicated the location of pStat3 expressing cells. The experiments were repeated at least two times. Bar: 200 μm. n=12, and 10/12 showed increased expression of pStat3. (C) Quantification of panel B. (D) Bar graph showing the relative calcium signals at the indicated conditions. Ovarian explants were cultured at the indicated conditions, and calcium signals were determined by calcium indicator dye Cal-520 acetoxymethyl ester. (E) Representative WB images showing the expression of the indicated makers in ovaries, early ovotestes, and middle ovotestes. n=5 per group. (F) Quantification of panel E for relative expression of Amh and pStat3. (G) pStat3 levels after the addition of A23187 and BAPTA-AM in cultured ovarian cells at 25 °C and 34 °C conditions. (H) Quantification of panel G. All experiments were repeated at least two times.

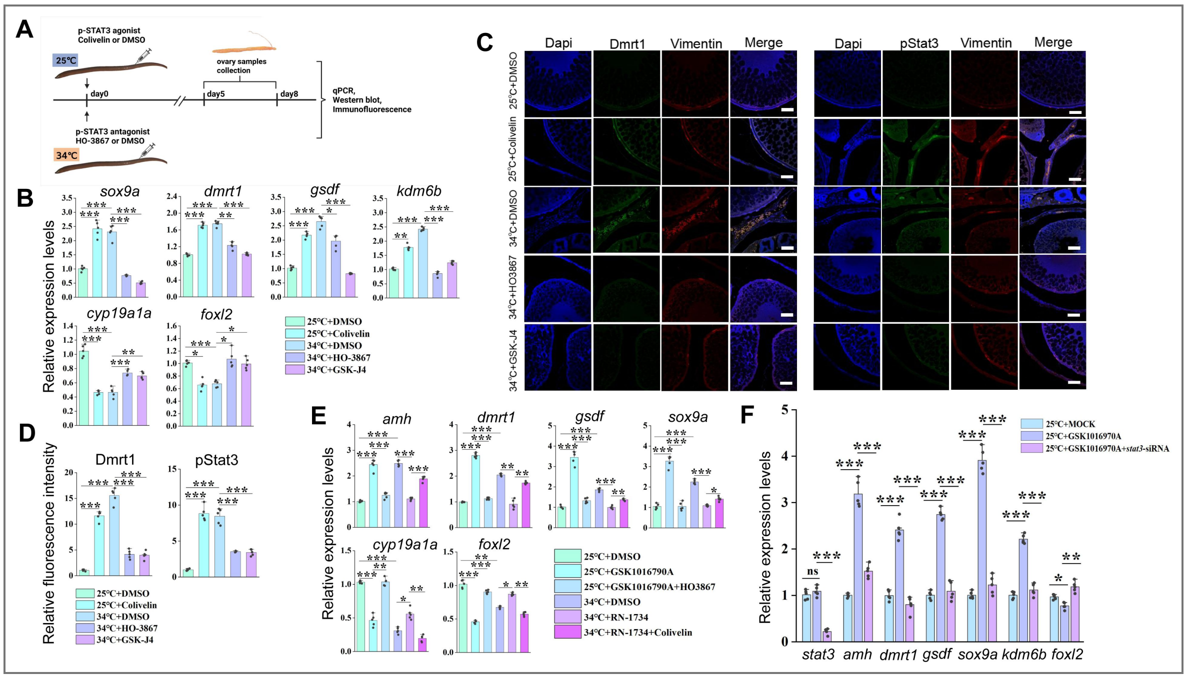


Figure 6. **Animal experiments showing that warm temperature induced male gene expression depends on pStat3.**

(A) Cartoon showing the design of the experiments. Female eels kept at cool (25 °C) and warm (34 °C) temperatures were injected with the pStat3 agonist Colivelin and antagonist HO3867 into the ovaries, respectively. After 2-3 days of injection, the ovaries were isolated and processed for the subsequent experiments. n=50. (B) qPCR results showing the relative expression of the indicated genes at the indicated conditions. n=5 per group. (C) IF images showing the expression of male biased genes at the indicated conditions. Bar: 200 µm. n=10, and 8/10 showed increased expression of pStat3/Dmrt1. (D) Quantification of panel C. n=5 per group. (E) qPCR results showing the relative expression of the indicated genes at the indicated conditions. n=5 per group. (F) qPCR results showing the relative expression of the indicated genes at the indicated conditions. n=5 per group. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. The qPCR experiments were repeated at least three times.

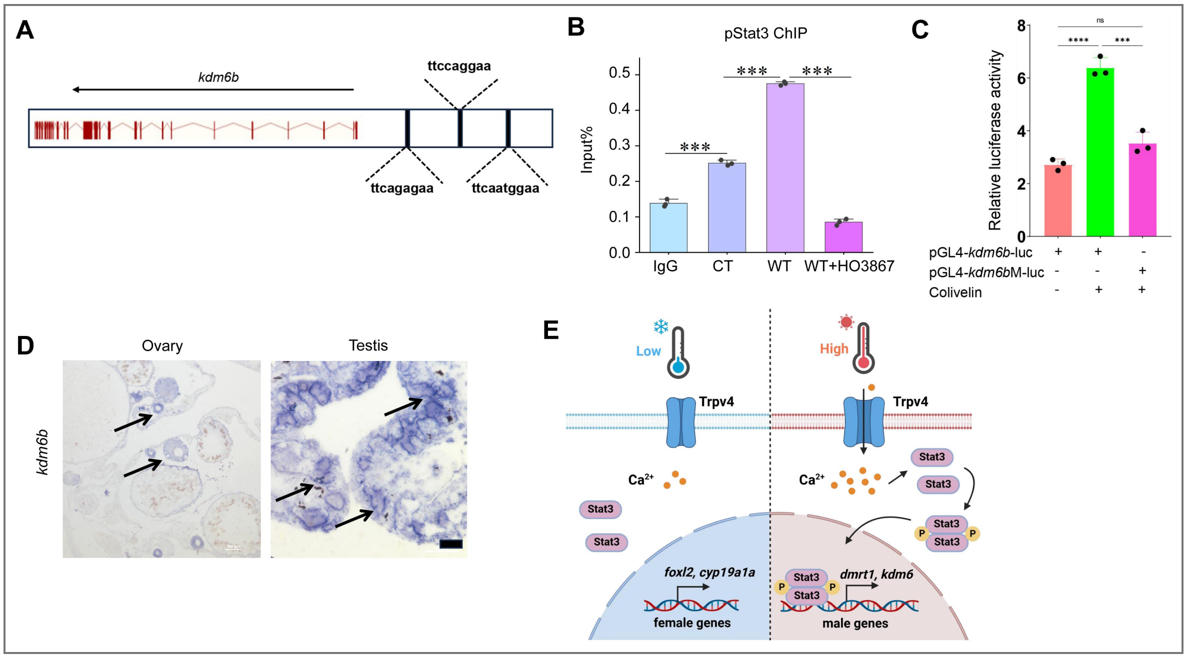


Figure 7. **pStat3 binds and activates the *kdm6b* gene.**

(A) Cartoon showing the conserved pStat3 binding motifs upstream the TSS of the *kdm6b* gene. (B) ChIP experiments showing the enrichment of pStat3 at the *kdm6b* locus in ovarian tissues of fish reared at cool temperature (CT) and warm temperature (WT) conditions, in the absence and presence of HO-3867. (C) Luciferase assay for *kdm6b*-luc and *kdm6b*M-luc activities in 293T cells, in the absence and presence of pStat3 agonist Colivelin. (D) The representative ISH images showing the expression of *kdm6b* in ovary and testis. Bar: 200 μm. n=3 per group. (E) Cartoon showing how Trpv4 may link environmental temperature to the sex determination cascades via the downstream signaling pathways in ricefield eel. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. ChIP and Luciferase experiments were repeated two times.

of GSK1016790A or Colivelin at 25 °C, and was down-regulated by the injection or addition of RN1734 or HO-3867 at 34 °C (Figure 3B; Figure 6B; Figure S3B; Figure S5A). Thus, the *kdm6b* gene displayed an expression pattern similar to that of the testicular differentiation genes, which strongly supported that *kdm6b* is a male pathway gene downstream of Trpv4-Ca^{2+}-pStat3.

If *kdm6b* is downstream of Trpv4-pStat3 to regulate the expression of *dmrt1*, inhibition Kdm6b demethyltransferase activity should prevent up-regulation of testicular differentiation genes by warm temperature treatment, similar to HO-3867 and RN1734 treatment. 0.5 μM GSK-J4, an Kdm6b specific inhibitor, was injected into the ovaries of ricefield eel, and the expression of sex-biased factors were examined by the qPCR and IF analyses. The results showed that GSK-J4 injection significantly down-regulated the expression of testicular differentiation factors at the expense of ovarian differentiation factors (Figures 6B-D). Similar results were obtained in cultured ovarian cells (Figures S5A). We propose that there exists a Trpv4-pStat3-Kdm6b axis that links the environmental temperature and the male sex determination pathway (Figure 7E).

Discussion

Since the first discover that ricefield eel is a teleost fish of hermaphroditism (Liu et al. 1944), the underlying mechanism has been under intensive investigation. However, it is still mysterious, partially because it is challenging to perform genetic studies due to unique reproductive strategy and life history of the species (Song et al., 2022). In this work, we provided solid evidences that warm temperature promotes the sex change of adult ricefield eel, and that the ion channel protein Trpv4 functions as an important molecular linker connecting the environmental temperature and the sex determination pathway. Our results support a model in which temperature-driven sex change is achieved via the Trpv4-pStat3-Kdm6b-Dmrt1 axis. In this model, warm temperature exposure leads to increased calcium influx via Trpv4, which promotes phosphorylation of pStat3 and expression of the male pathway genes such as *kdm6b/dmrt1* in the ovary, eventually and gradually resulting in transformation of ovary to testis via an ovotestis. Our work for the first time provides the mechanistic explanation of how natural sex change occurs in aging ricefield eel, which may serve as a paradigm to study natural sex change of other adult animals, including the hermaphroditic marine fish.

The initial perception and translation of environmental cues into the sex determination cascades remain unclear in any species. Previous studies have proposed endoplasmic reticulum chaperone, heat shock proteins and transmembrane ion channels, as potential temperature sensors (He et al., 2010; Shi et al., 2024). In contrast to Cdc2-like kinases (Clks) that response to subtle temperature change in homeothermic organisms, the transient receptor potential (TRP) cation channels may play important roles in poikilothermic animals that are exposed to strong ambient temperature change (Haltenhof et al., 2020; Huang et al., 2024). The TRP channel family can be divided into at least seven subfamilies, including TRPA, TRPC, TRPM, TRPML, TRPN, TRPP and TRPV. The thermo-TRPs contain nine members, such as TRPV (TRPV1-4). TRPV1-3 are mainly activated by temperatures above 35 °C, respectively (Mo et al., 2022). Of note, TRPV4 is activated by warm temperatures (27-35 °C) (Goikoetxea et al., 2021; Güler et al., 2002; Fujita et al., 2017), which are physiologically relevant to the spawning and/or initiation of sex change of ricefield eel. Trpv4 has been shown to be abundantly expressed during testicular/sperm development (Kumar et al., 2016; Mundt et al., 2018). Activation of Trpv4 by various endogenous/exodogenous stimuli increases Ca^{2+} influx, participating in multiple downstream biological events, including gonadal cell apoptosis (Guler et al., 2002; Liu et al., 2021; Luo et al., 2023; Vrenken et al., 2016; Yamamoto et al., 2024). Importantly, Trpv4 has been shown to associate temperature and sex determination in TSD alligator, and pharmaceutical activation and inhibition of Trpv4 alters testis differentiation (Yatsu et al., 2015). In ricefield eel, *trpv4* was one of the most up-regulated *trp* genes by warm temperature treatment, and its expression was associated with testicular development. Thus, we focused on the role of Trpv4 in this work. Based on in vitro and in vivo experiments, we demonstrated that a portion of Trpv4 expressing gonadal somatic cells are highly sensitive to temperature cues and are functionally important in temperature-driven sex change of ricefield eel.

We provided evidences that the pStat3 signaling is downstream of and mediates Trpv4. Previous work in the red-eared slider turtle (*Trachemys scripta elegans*) and the bluehead wrasse (*Thalassoma bifasciatum*) has indicated that Stat3 phosphorylation and epigenetic regulation are involved in TSD or sex change (Todd et al., 2019). pStat3 is likely to be involved in sex determination by transcriptionally activating the female pathway genes such as *foxl2* and/or repressing the male pathways genes such as *kdm6b* (Chen et al., 2024; Deveson et al., 2017; Ge et al., 2018; Holleley et al., 2015; Weber et al., 2020; Wu et al., 2024). The function of *jmjd3/kdm6b* in male sex determination in embryonic TISR/TSD animals has been well established (Chen et al., 2024; Ge et al., 2018; Yao et al., 2023). Overexpression and knockdown of *kdm6b* cause female to male or male to female sex reversal, respectively. In this work, increased expression of testicular differentiation genes preceded the down-regulation of ovarian differentiation genes, suggesting that the male pathway genes were more directly regulated during sex change of ricefield eel. Our ChIP and luciferase reporter experiments suggested that pStat3 directly binds and activates the *kdm6b* gene. Whether pStat3 regulates female genes such as *foxl2* awaits further investigation. In turtles, pStat3 functions as a repressor of *kdm6b* (Weber et al., 2020), whereas in ricefield eel and the bluehead wrasse, pStat3 may function as an activator of *kdm6b* (Todd et al., 2019). We propose that a yet-unidentified co-factor may determine whether pStat3 is a transcriptional repressor or activator. Nevertheless, a conserved Trpv4-pStat3-Kdm6 axis may play an important role in sex determination/differentiation in TSD/GSD+ TE reptiles and fish. In the future, it will be interesting to investigate whether this is the case using other TISR/TSD models.

Trpv4, pStat3 and Kdm6b were expressed in somatic cells surrounding ovarian oocytes and interstitial cells. After warm temperature exposure, their levels were elevated over time, similar to that of the testicular differentiation genes. We propose that the magnitude and duration of temperature exposure promote sex change of ricefield eel by driving the accumulation of testicular differentiation genes in sufficient quantities (Weber et al., 2020). In sex reversal animals, a long existing question is what cells in the gonad respond to environmental temperatures to initiate sex change. In groupers such as *Epinephelus akaara*, certain pre-existing somatic cells, called testicular-inducing steroidogenic cells, trigger the sex change (Murata et al., 2021). Recently, it has been proposed that a group of Nr5a1^{+}Trp^{-} thermosensitive steroidogenic cells use temperature-dependent Ca^{2+} signals to transduce into the male sex determination pathway in turtles (Li et al., 2025; Ye et al., 2025). Here, we showed that in ricefield eel, a portion of Trpv4-expressing granulosa cells in ovary are responding to temperature change, and may play an important role in triggering the sex reversal. Dmrt1 expression showed up within follicles in a typical granulosa cell location (Figure 3E; 6C). We propose that warm temperature exposure may cell-autonomously reprogram of a portion of Trpv4^{+} granulosa cells into Dmrt1/Sox9a-positive Sertoli precursor cells, thereby promoting the sex change of ricefield eel.

To summary, this study used ricefield eel hermaphrodite to elucidate the molecular basis underlying the transduction of environmental temperature into an intracellular signal for sex determination. We have made a few important findings. First, the ovarian somatic cells in ricefield eel are highly responding to the warm temperature. Warm temperature is sufficient to induce the expression of testicular differentiation genes, without input of other factors such as hormones. Second, we identified a group of Trpv4-expressing granulosa cells that can directly perceive and respond to ambient temperature changes. Third, the pStat3-Kdm6b-*dmrt1* axis is downstream and mediates temperature-evoked Trpv4 activation. Our work revealed a comprehensive TISR mechanism, which involves signals that initiate sex reversal (temperature) and the capture (Trpv4), sensing (Trpv4-controlled calcium influx), transduction and interpretation (the Jak/Stat3 pathway) of environmental signals into the sex determination pathway (*kdm6b/dmrt1*). Of course, this work has limitations. Due to the unique life history of ricefield eel, genetic evidences are not sufficiently provided to substantiate the conclusion. Direct evidences for Trpv4 control of Ca^{2+} signaling are lacking. Whether and how Trpv4-positive granulosa cells transdifferentiate into Dmrt1-positive Sertoli precursors during sex change awaits further investigation.

Materials and Methods

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Yuhua Sun (*sunyh@ihb.ac.cn*).

Materials availability

All antibodies and plasmids generated in this study are available from the lead contact.

Sampling and maintenance of ricefield eel

Ricefield eel were purchased from the Baishazhou market, Wuhan, China. They were temporarily maintained in the lab at 25±1 °C under a 14-h light/10-h dark cycle, and fed daily with commercial diet. Female fish were usually less than 40 cm in length, males were more than 50 cm in length, and intersex animals were of medium length.

Animal experiments and treatments were performed according to the Guide for Animal Care and Use Committee of the Institute of Hydrobiology, Chinese Academy of Sciences (IHB, CAS, Protocol No. 2016-018).

RNA preparation and RNA-sequencing

We have performed RNA-sequencing (RNA-seq) experiments using gonadal tissues from female, middle-late intersex, and male animals (Zhang et al., 2025). To explore the earliest events that trigger the onset of sex change of ricefield eel, gonads from female, early intersex, and middle intersex animals were isolated. Each gonad was examined by color and morphology, and in some cases histological experiments were performed to confirm the gonadal identities. Two to three gonads from each group were pooled, and total RNAs were isolated using a TRIzol reagent (ThermoScientific, USA). RNA sample quality was checked by the OD 260/280 ratio using a Nanodrop 2000. The total mRNAs were sent to the BGI (Beijing Genomics Institute) company (ShenZhen, China), where RNA-sequencing libraries were constructed and sequenced by a BGI-500 system. RNA-seq experiments were performed at least two times, with two technical repeats.

Quantitative real-time PCR (qPCR) experiments

Total RNA was isolated using the Isolation Kit mRNA. *dmrt1/sox9a* were used as male-specific genes, and *cyp19a1a/foxl2* were used as female-specific genes. The primers and the related information used in this work were listed in Table S2. qPCR analysis was used to determine gene expression levels. A total of 1 μg RNA was reverse transcribed into cDNAs using the TransScript All-in-One First-Strand cDNA synthesis Supermix (Transgen Biotech, China, AT341). qPCR amplification was carried out on a Bio-Rad CFX96 Touch Real-Time PCR System (Bio-Rad, Hercules, CA, USA) in triplicate. The reaction mixture consisted of 5 μL PerfectStart™ Green qPCR SuperMix (Transgen Biotech, China), 3.6 μL ddH2O, 0.2 μL forward and reverse primers, and 1 μL cDNA. The cycling parameters used were 94 °C for 30 s, 94 °C for 5 s, 60 °C for 15 s, and 72 °C for 10 s for 40 cycles. Quantification cycle or cycle threshold values were determined using CFX Manager 3.1 (Bio-Rad, USA). Primers were 20-21 nucleotides long, with a melting temperature between 58 and 60 °C and a guanine-cytosine content between 50% and 60% generating an amplicon of 80-200 bp. Beta-actin was used as a reference gene. All qPCR experiments were repeated three times, and the relative gene expression levels were calculated based on the $2^{-\Delta\Delta Ct}$ method.

siRNA knockdown experiments

To study in vivo function of Trpv4, we used siRNA to deplete the expression of the *trpv4* gene. The sequences of the ricefield eel *trpv4* gene (Accession Number: NW_018127903.1) were obtained from NCBI GenBank. The siRNA sequences are list in Table S2. These siRNAs were purchased from Sangon Biotech (Shanghai, China). Eighteen female fish were equally divided into 3 groups:

25 °C+ MOCK; 34 °C+ MOCK; 34 °C+ *trpv4*-siRNA. For the 34 °C-group setting, after 1-2 day of acclimation at 25 °C, the temperature of water was gradually increased by 3 °C per day until reaching to 34 °C. For RNAi experiment, before increasing the temperatures, the siRNA was injected into ovaries through genital papilla. siRNAs were injected twice, for an interval of 2 days. Three *trpv4*-siRNAs were mixed in equal amounts; each fish was injected with 100 nM/kg. The MOCK groups were injected with equal amounts of control siRNAs. Two days after the second siRNA injection, ovarian samples were processed for qPCR analysis, and/or cryopreserved for ISH.

WB analysis

WB was performed as previously described (Sun et al., 2020; 2023). The antibodies used in this work were: Amh (Huabio, #HA500137, China), Dmrt1 (home-made), Sox9a (home-made), Foxl2 (Thermofisher, #PA1-802, USA), Stat3 (Cell signaling, #9139, USA), pStat3 (Cell Signaling, #9145, USA). For animal experiments, the fish were reared for additional 3-5 days for the detection of protein of interest. To validate the specificity of the antibodies, siRNA-mediated knockdown with immunoblot quantification with at least 2 replicates were performed.

IF experiments

Anesthetic fish were fixed with 4% PFA, and gonads were isolated. The gonads were washed three times with PBS and dehydrated in sucrose solution (15% sucrose/PBS, 30% sucrose/PBS) for 2 h at 4 °C. Gonads were mounted in Tissue-Tek OCT compound (#4583, Sakura) and sectioned to 30 μm thickness on a cryostat. Slides containing sections were dried at room temperature for 30 min and washed 3 times for 5 min at room temperature with PBS. Sections on slides were blocked using 5% normal bovine serum (#A2153, Sigma Life Science) in PBS + 0.1% Triton-X100 (#V900502, VETEC) for 2 h. After wash, Primary antibodies, including Amh (Huabio, China), Foxl2 (ab5096, abcam), Trpv4 (Huabio, ER65407, China), Dmrt1 (home-made), Sox9a (home-made), Vimentin (OMA1-06001, Thermofisher, USA), were added at dilution of 1: 1000. Slides were washed 3 times with PBS for 5 min, followed by incubation in Alexa 488 or Alexa 555 (#A11008/A21428, Thermofisher, USA) secondary antibodies (1: 500) for 2 h. The samples were counter-stained with DAPI (#D9542, Sigma, 1:1000) in 1×PBS at room temperature for 1 h. After 3 times washing with PBS, slides were mounted using an anti-fade mounting medium (#HY-K1042, MedChemExpress, China). Mounted slides were imaged with a Leica Confocal Microscope (TCS SP8 STED, Germany).

In situ hybridization (ISH)

To detect the expression of *trpv4/kdm6b* in the gonads, ISH experiments were performed. The cDNAs of ricefield eel *trpv4* and *kdm6b* were amplified by gene specific primers *trpv4*-SP6-F: ATTT AGGTGACACTATAGAAGCGTTTCTAGCCATTTCCTATCGT, and *trpv4*-T7-R: TAATACGACTCACTATAGGG AGACATTATCTGCTCCTAATCGAACC.

All other primers can be found in Table S1. Digoxin labeled RNA probes of Sp6-sense and T7-antisense were synthesized using the DIG RNA Labeling Kit Sp6/T7 (Roche, Basel, Switzerland). ISH was conducted following the methodology outlined below. Briefly, the fixed gonads were processed by dehydration, paraffin embedding and serial sectioning (5 μm). Then the gonad slices were digested at 37 °C with 200 ng/mL proteinase K for 5 min. Hybridization was carried out for 16 h at 60 °C using a probe concentration of 1 ng/μL in the hybridization buffer. The samples were incubated with the Anti-Digoxigenin-AP conjugate (Roche, Basel, Switzerland) at a 1: 2500 dilution for 16h at 4 °C, and stained in NBT/BCIP staining solution (Roche, Basel, Switzerland) in the dark for 0.5–1.5h at room temperature. The results were observed and photographed using an optical microscope (Zeiss, Oberkochen, Germany). Drawings and final panels were designed using Adobe Photoshop CS6 (San Jose, CA, USA).

ChIP experiments

ChIP experiments were performed according to the Agilent Mammalian ChIP-on-chip manual. Briefly, gonadal tissues were processed into single cells and were fixed with 1% formaldehyde for 10 min at room temperature. The reactions were stopped by 0.125 M Glycine for 5 min with rotating. The fixed chromatin was sonicated to an average of (500-1000) bp (for ChIP qPCR) using the S2 Covaris Sonication System (USA) according to the manual. Then Triton X-100 was added to the sonicated chromatin solutions to a final concentration of 0.1%. After centrifugation, 50 μL of supernatants were saved as input. The remainder of the chromatin solution was incubated with Dynabeads previously coupled with 5 μg ChIP grade pStat3 antibodies overnight at 4 °C with rotation. The next day, after 7 times washing with the wash buffer, the complexes were reverse cross-linked overnight at 65 °C. DNAs were extracted by hydroxybenzene-chloroform-isoamyl alcohol and purified by a Phase Lock Gel (Tiangen, China). The ChIPed DNAs were dissolved in 100 μL distilled water. qPCR was performed using a Bio-Rad instrument. The enrichment was calculated relative to the amount of input as described. All experiments were repeated at least two times. The relative gene expression levels were calculated based on the $2^{-\Delta\Delta Ct}$ method. The paired t-test was used for the statistical analysis. Data were shown as means± SD.

Luciferase Assay

HEK293T cells were seeded in 24-well plates in DMEM medium containing 10% FBS for 24h. The cells were then transiently transfected with the pGL4-*kdm6b*-luc or pGL4-*kdm6b*M-luc reporters using Lipofectamine 2000 (Invitrogen). pTKRenilla was used as an internal control. The luciferase activity was measured with the Dual-luciferase Reporter Assay system (Promega).

Hematoxylin and eosin (H&E) experiments

H&E experiments were used for the identification of gonadal types in ricefield eels. The gonads were fixed in Bouin's solution for at least 24 h, and the H&E experiments were performed by Wuhan Icongene Biotechnology Company. Briefly, dehydration and paraffin imbedding were then performed on the ASP6025S Automatic Vacuum Tissue Processor (Leica, Wetzlar, Germany). The samples were sectioned using the Leica microtome (Leica) at a thickness of 5 μm. After de-paraffinization, hydration and staining, the sections were examined on the Nikon ECLIPSE Ni-U microscope and micrographs were taken with the Digit Sight DS-Fi2 digital camera (Nikon).

Gonadal sex identification

The gonadal types were initially identified according to morphological features including the size, shape and color. The gonadal sex of each fish was confirmed by histological sectioning and microscopic observation. In some cases, gene expression analysis was also used to confirm the gonadal sex types. Male genes such as *dmrt1* were not expressed in ovaries, slightly up-regulated in early ovotestes, and abundantly expressed in middle– and late-ovotestes.

Long term temperature experiments

The aim of this experiment was to assess the gonadal phenotypes of female fish that were reared at 25 °C (cool temperature, CT) vs 33/34 °C (warm temperatures, WT) over 6 months. The experiments were performed from September-3, 2024. 1.5-year-old wild female fish were transiently maintained for 3-5 day at the laboratory, and unhealthy animals were discarded. A total of approximately 400 female fish were randomly divided into the CT and WT groups, stocked in 10-12 tanks, at a density of 15-20 fish per tank. Fish were fed with commercial diet.

1, 3, 6 months later, one tank in each group was randomly selected, and fish were anaesthetized and measured for body length and weight. The gonads were isolated and subjected to the histological analysis to determine the gonadal sex types. In some cases, gene expression analysis was used to determine the gonadal sex.

Short term temperature experiments Animal experiments

The experiments were used to explore how warm temperature treatment affects the expression of sex differentiation genes in ovaries in a short period of time (3-10 days). The ricefield eels were reared at 25 °C (cool temperature) and at 34 °C (warm temperature). The temperature-increase protocol started at 25 °C, with a progressive increase of 3 °C per day until reaching 34 °C. Temperature was monitored twice a day throughout the experiment. Fish were fed with Artemia daily.

The ricefield eels from the Baishazhou market were transiently raised for 1-2 day at cool temperature (25 °C). The fish were then divided into 4 groups based on the temperature and the injected small molecules: 25 °C+ DMSO; 25 °C+ GSK1016790A; 34 °C+ DMSO; 34 °C+ RN1734. For the 34 °C group setting, the temperature of water was gradually increased by 3 °C every day until reaching to 34 °C. Before increasing the temperatures, the small molecules of appropriate doses were injected into ovaries. The final concentrations used were at 0.02 mg/kg body weight for RN1734, 0.01 mg/kg body weight for HO-3867 or GSK1016790A or Colivelin, and a similar volume of 1‰ DMSO were injected and served as control.

To determine the upstream and downstream relationships between Trpv4 and pStat3, rescue experiments were performed by injecting the small molecules into the ovaries. Six groups were set up based on the temperature and the injected small molecules: 25 °C+ DMSO; 25 °C+ GSK1016790A; 25 °C+ GSK1016790A+ HO3867; 34 °C+ DMSO; 34 °C+ RN1734; 34 °C+ RN1734+ Colivelin.

To investigate the role of Kdm6b, 0.5 μM GSK-J4, an Kdm6b specific inhibitor, was injected into the ovaries or added into the cultured ovarian explants and/or cells.

Ovarian explant or cell culture

The ovaries were isolated from female ricefields and washed with cold 2% pen/strep PBS 3 times. The ovaries were cut into 2 mm^3 pieces, and/or were digested by 0.25% TrypE for 30 min into single cells. For single cell culture, after filtration, ovarian cells were plated and cultured in 12 well plates.

For pharmaceutical experiments for Trpv4, the cells were divided into 4 groups: 26 °C+ DMSO; 26 °C+ GSK1016790A; 33 °C+ DMSO; 33 °C+ RN1734. For the 33 °C group setting, after 1-2 day of acclimation at 25 °C, the temperature of water was gradually increased by 3 °C every day until reaching to 33 °C. The doses of small molecules were optimized, and the final concentrations used were at 10 μM for RN1734, 100 nM for GSK1016790A and a similar volume of 1‰ DMSO were added and served as control. For pharmaceutical experiments for pStat3 function, 2 μM HO3867 and 20 μM Colivelin were added.

To determine the upstream and downstream relationships between Trpv4 and pStat3, rescue experiments were performed. Six groups were set up based on the temperature and the small molecules: 25 °C+ DMSO; 25 °C+ GSK1016790A; 25 °C+ GSK1016790A+ HO3867; 33 °C+ DMSO; 33 °C+ RN1734; 33 °C+ RN1734+ Colivelin.

Statistic analysis

For gene expression analyses, differences in mean values between two groups were assessed using Student's t-test. A one-way ANOVA was used to compare the expression levels of each target and the differences were determined using Tukey's post hoc test. Significance was defined as *p <0.05, **p <0.01, ***p <0.001, ****p <0.0001. Data are presented as mean standard error (SEM). Statistical analyses were conducted and graphs.

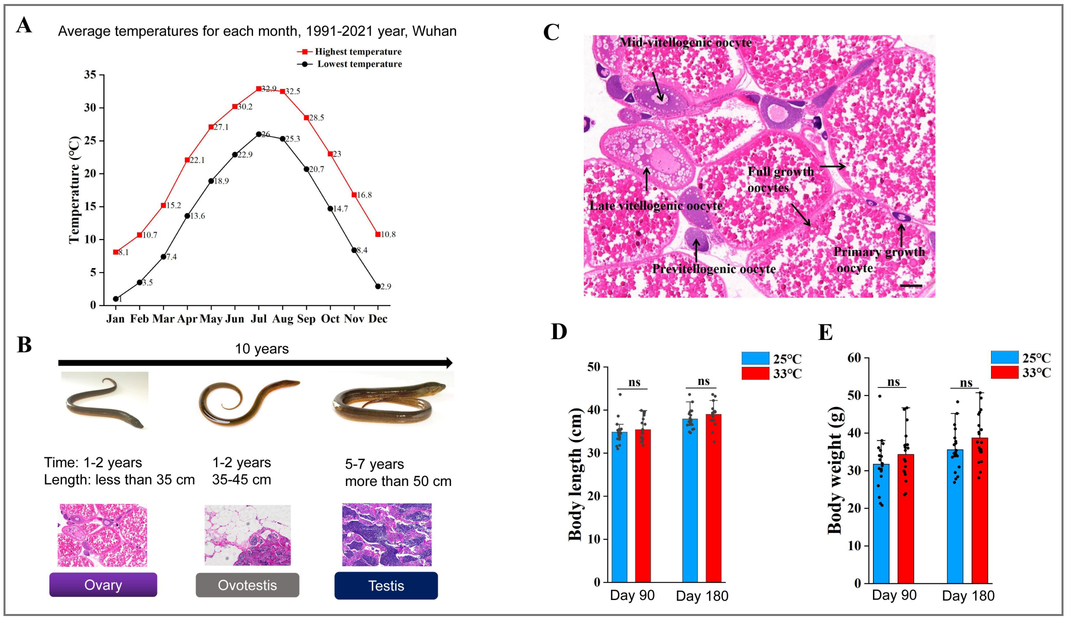
Supplementary Figure legends

Acknowledgements

This work was supported by the National Key Research and Development Program of China (2022YFD2400101) to YH Sun. We thank Tanhong Eel Industry Aquaculture Co., Ltd (Chibi city, Hubei Province) for collecting the fish materials and fish framing. We are grateful to Prof. Jianzhen Li from Northwest Normal University for critical comments for this work.

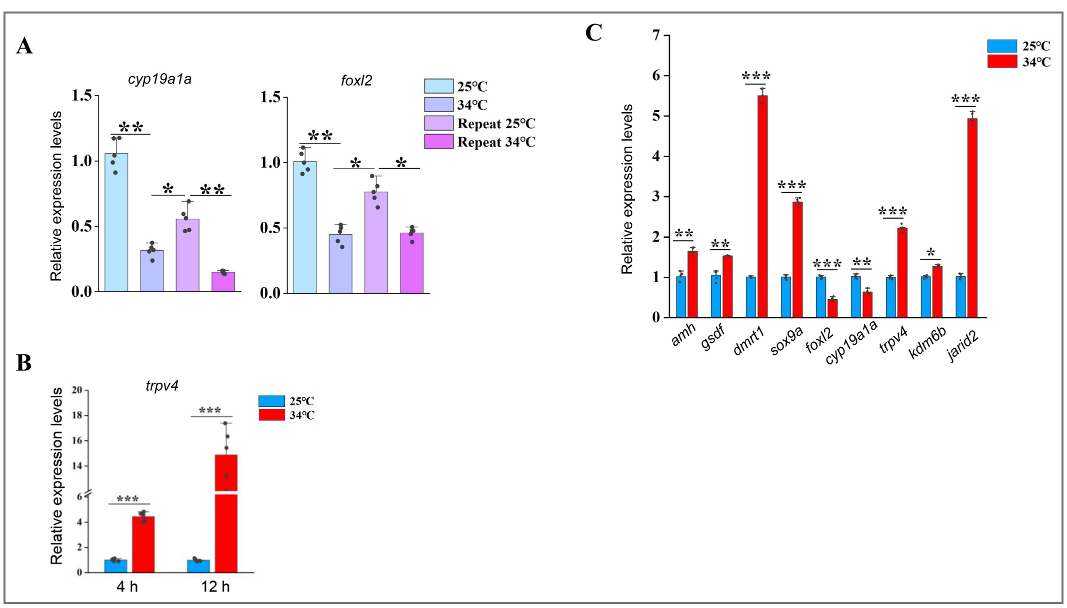
Supplementary Figure 1. Related to Figure 1.

(A) The graph showing the annual temperature dynamic by month for 30 years in Wuhan area, Hubei province, China. The highest and lowest temperatures per month were shown. (B) Schematic of the whole process of sex change, summarizing morphology, length and gonadal histology across time (10 years). The ricefield eel are born as females. After 2 year growth, they reach sexual maturity. After spawning (ranging from May to August each year), females will enter 1-2 years of intersex stage before becoming functional males. The average lifespan of wild ricefield eel is around 10 years. (C) H&E staining images showing the typical cell types in an ovary of a 2-year-old female, before spawning. Bar: 200 µm. (D) The average body length of fish that were raised for 90 and 180 days at the indicated temperatures. (E) The average body weight of fish that were raised for 90 and 180 days at the indicated temperatures. ns: not significant. The experiments were repeated two times.



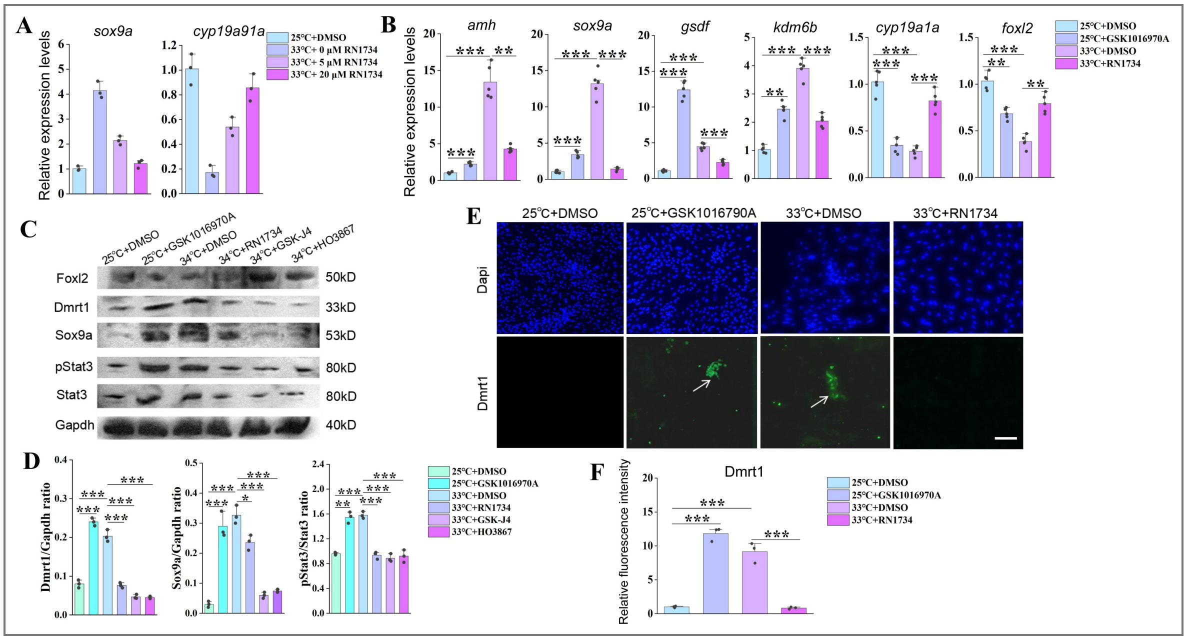
Supplementary Figure 2. Related to Figure 2.

(A) The qPCR results showing the expression patterns of the indicated female sex genes in repeated temperature shifting experiments of in vitro cultured ovaries. n=5 per group. (B) The qPCR results showing the expression of *trpv4* at the indicated time points of in vitro cultured ovarian explants. n=5 per group. (C) The qPCR results showing the expression of the indicated temperature responding– and sex-biased genes in ovaries of females that reared at 25 °C and after 1 day exposure to 34 °C. n=5 per group. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. All experiments were repeated at least three times.



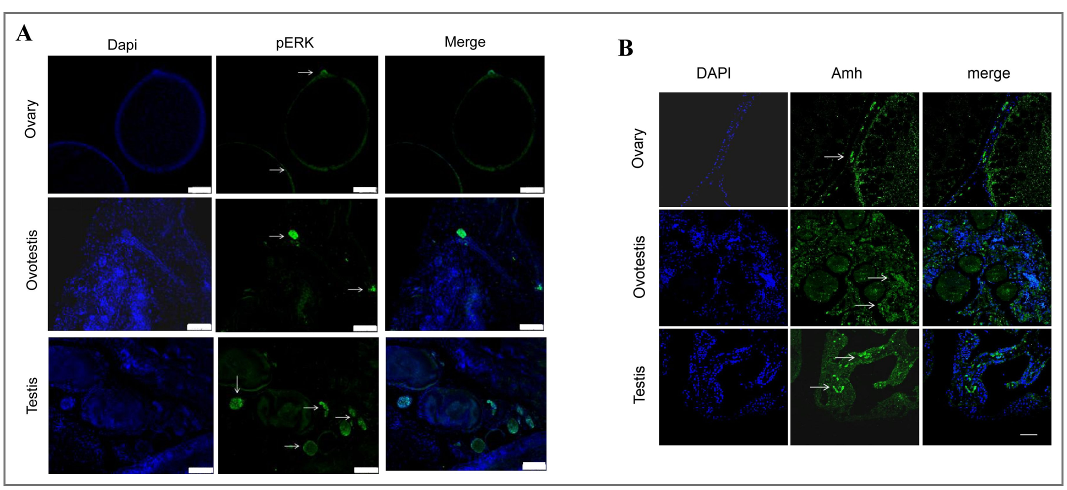
Supplementary Figure 3. Related to Figure 3.

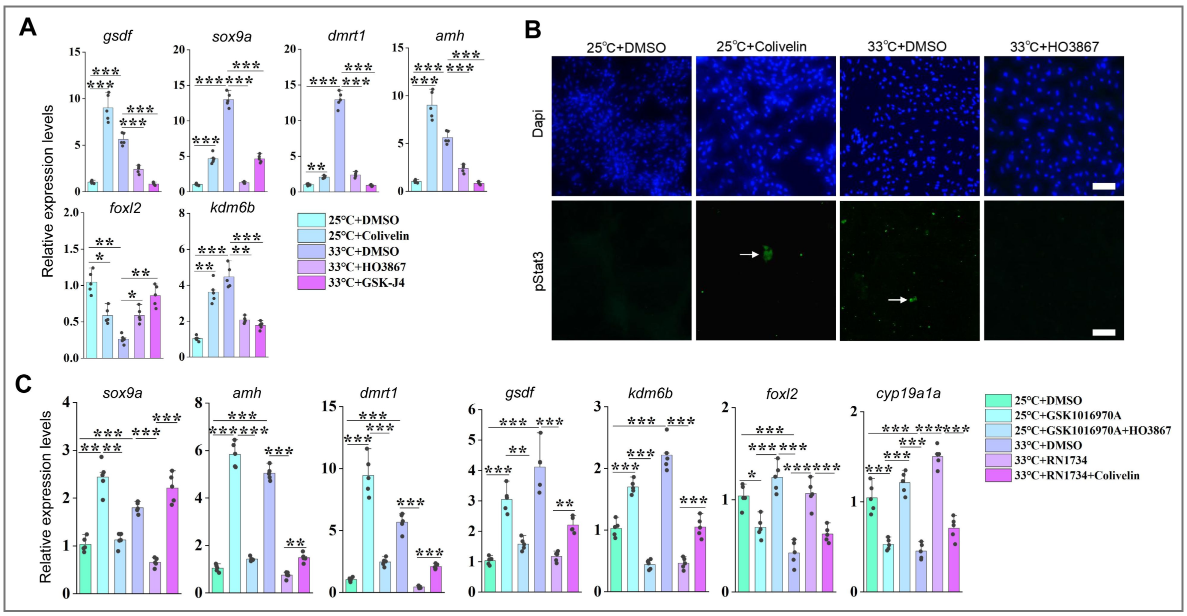
(A) qPCR results showing the expression of the indicated male– and female-biased genes in ovarian explants cultured at 25 °C and 34 °C with increasing doses of small molecule RN1734. n=5 per group. (B) qPCR results showing the expression of the sex-biased genes at the indicated conditions, based on in vitro cultured ovarian explants. n=5 per group. (C) WB images showing the expression levels of the indicated proteins at the indicated conditions, based on in vitro cultured ovarian explants. (D) Quantification of panel C. (E) IF images showing the expression levels of Dmrt1 at the indicated conditions, based on in vitro cultured ovarian explants. n=5 per group. (F) Quantification of panel C. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. All experiments were repeated at least three times.



Supplementary Figure 4. Related to Figure 5.

(A-B) The representative IF images showing the expression of pERK and Amh from ovaries, ovotestes, and testes. n=6 per group. The experiments were repeated at least two times.





Supplementary Figure 5. **Related to Figure 6.**

(A) qPCR results showing the expression of the indicated genes at the indicated conditions, based on in vitro cultured ovarian explants. (B) Representative IF images showing the expression of pStat3 at the indicated conditions, based on in vitro cultured ovarian explants. n=6 per group. (C) qPCR results showing the expression of the indicated genes at the indicated conditions, based on in vitro cultured ovarian explants. *: *P*< 0.05, **: *P*< 0.01, ***: *P*< 0.001, and ****: *P*< 0.0001. ns: not significant. All experiments were repeated at least three times.

Additional information

Data availability

All data are available upon reasonable request

Author contributions

Z Yang, TT Luo and YM Zhang performed experiments and generated data; YH Sun planned and designed experiments; YH Sun wrote the manuscript; all authors edited the manuscript.

Funding

Funder	Grant reference number	Author
MOST | National Key Research and Development Program of China (NKPs)		Yuhua Sun

Additional files

Supplementary file 1. Uncropped western blot figures.

Table S1.

Table S2.

References

1. **Castelli M.A.,** Whiteley S.L., Georges A., Holleley C.E (2020) Cellular calcium and redox regulation: the mediator of vertebrate environmental sex determination?. *Biol. Rev. Camb. Phil. Soc* **95**:680-695 https://doi.org/10.1111/brv.12582
2. **Chan S.H,** Phillips J.G (1967) Structure of gonad during natural sex reversal in *Monopterus albus* (Pisces: Teleostei). *Journal of Zoology* **151**:129-141
3. **Chen Q.,** Yang D., Chen M., Xiong J., Huang J., Ding W., Gao K., Lai B., Zheng L., Tang Z., *et al.* (2024) Smad4 and FoxH1 potentially interact to regulate *cyp19a1a* promoter in the ovary of ricefield eel (*Monopterus albus*). *Biol. SexDiffer* **15**:60-71
4. **Cheng H.,** Zhou R (2022) Decoding genome recombination and sex reversal. *Trends Endocrinol. Metabol* **33**:175-185
5. **Deveson I.W.,** Holleley C.E., Blackburn J., Marshall Graves J.A., Mattick J.S., Waters P.D., Georges A (2017) Differential intron retention in Jumonji chromatin modifier genes is implicated in reptile temperature-dependent sex determination. *Sci. Adv* **3**:e1700731
6. **Dupont S.,** Garcia-Moreno A., O'Neil E V, Capel B (2025) KDM6B is a conserved activator at the top of the male sex determination pathway. *Development* **152**:dev204724
7. **Fan M.,** Yang W., Zhang W., Zhang L (2022) The ontogenic gonadal transcriptomes provide insights into sex change in the ricefield eel *Monopterus albus*. *BMC Zool* **7**:56-71
8. **Fan M.,** Yang W., Sun S., Li Z., Zhang L.H., Zhang W (2021) Natural sex change of the "virgin" female ricefield eel Monopterus albus. *Acta Hydrobiologica Sinica* **45**:387-396 https://doi.org/10.7541/2021.2020.142
9. **Ge C.T.,** Ye J., Weber C., Sun W., Zhang H.Y., Zhou Y.J., Cai C., Qian G.Y., Capel B (2018) The histone demethylase KDM6B regulates temperature-dependent sex determination in a turtle species. *Science* **360**:645-650
10. **Güler A.D.,** Lee H., Iida T., Shimizu I., Tominaga M., Caterina M (2002) Heat-evoked activation of the ion channel TRPV4. *J Neurosci* **22**:6408-6414

11. **Haltenhof T.,** De Bortoli F., Schiefer S., Meinke S., Emmeriches A.K., Petermann K., Timmermann B., Imhof P., Franz A., Loll B., *et al.* (2020) A conserved kinase-based body-temperature sensor globally controls alternative splicing and gene expression. *Molecular Cell* **78**:57-69

12. **He Y.,** Shang X., Sun J., Zhang L., Zhao W., Tian Y., Cheng H., Zhou R (2010) Gonadal apoptosis during sex reversal of the rice field eel: implications for an evolutionarily conserved role of the molecular chaperone heat shock protein 10. *J. Exp. Zool. B Mol. Dev. Evol* **314**:257-266 https://doi.org/10.1002/jez.b.21333

13. **He Z.,** Deng F., Yang D., He Z., Hu J., Ma Z., Zhang Q., He J., Ye L., Chen H., *et al.* (2022a) Crosstalk between sexrelated genes and apoptosis signaling reveals molecular insights into sex change in a protogynous hermaphroditic teleost fish, ricefield eel *Monopterus albus*. *Aquaculture* **552**:737918-737931 https://doi.org/10.1016/j.aquaculture.2022.737918

14. **He Z.,** Ye L., Yang D., Ma Z., Deng F., He Z., Hu J., Chen H., Zheng L., Pu Y., *et al.* (2022b) Identification, characterization and functional analysis of gonadal long noncoding RNAs in a protogynous hermaphroditic teleost fish, the ricefield eel (*Monopterus albus*). *BMC Genom* **23**:450-467

15. **He Z.,** Xiao F., Yang D., Deng F., Ding W., He Z., Wang S., Chen Q., Wang H., Chen M., *et al.* (2024) Protein expression patterns and metal metabolites in a protogynous hermaphrodite fish, the ricefield eel (*Monopterus albus*). *BMC Genom* **25**:500-515

16. **Holleley C.E.,** O'Meally D., Sarre S.D., Graves J.M., Ezaz T., Matsubara K., Azad B., Zhang X.W., Georges A. (2015) Sex reversal triggers the rapid transition from genetic to temperature dependent sex. *Nature* **523**:79-83

17. **Hu Q.M.,** Lian Z.T., Xia X.P., Tian H.F., Li Z (2022) Integrated chromatin accessibility and DNA methylation analysis to reveal the critical epigenetic modification and regulatory mechanism in gonadal differentiation of *Monopterus albus*. *Biology of Sex Differences* **13**:73-88

18. **Huang J.W.,** Wang X.Y., Gao X.Y., Liu Q.H., Li J (2024) Transient receptor potential (TRP) channels in Sebastes schlegelii: Genome-wide identification and ThermoTRP expression analysis under high-temperature. *Gene* **910**:148317

19. **Huang X.,** Guo Y.Q., Shui Y., Gao S., Yu H.S., Cheng H.H., Zhou R.J (2005) Multiple alternative splicing and differential expression of *dmrt1* during gonad transformation of the ricefield eel. *Biology of reproduction* **73**:1017-1024

20. **Ji F.Y.,** Yu Q.X., Liu J.D (2001) Micro-isolation of individual chromosome for gene localization in ricefield eel. *Chinese Science Bulletin* **22**:1894 https://doi.org/10.3321/j.issn:0023-074X.2001.22.012

21. **Jiang Y.J.,** Luo H.R., Hou M.X., Chen J., Tao B.B., Zhu Z.Y., Song Y.L., Hu W (2022) Aromatase inhibitor induces sex reversal in the protogynous hermaphroditic rice field eel (*Monopterus albus*). *Aquaculture* **551**:737960-737972

22. **Kumar A.** (2015) Regulation of TRP channels by steroids: Implications in physiology and diseases. *General and comparative endocrinology* **220**:23-32

23. **Li J.M.,** Ye Y.Z., Zheng D., Pei Z.D., Du W.G., Yang S.L (2025) Molecular basis for thermal switch of temperature-dependent sex determination in a turtle species. *Science Bulletin* **70**:3319-3323

24. **Liem K.F** (1963) Sex reversal as a natural process in the synbranchiform fish *Monopterus albus*. *Copeia* **2**:303-312

25. **Lin J.Q.,** Zhou Q., Yang H.Q., Fang L.M., Tang K.Y., Sun L., Wan Q.H., Fang S.G (2018) Molecular mechanism of temperature-dependent sex determination and differentiation in Chinese alligator revealed by developmental transcriptome profiling. *Sci. Bull* **63**:209-212 https://doi.org/10.1016/j.scib.2018.01.004

26. **Lin Q.H.,** Mei J., Li Z., Zhang H.M., Zhou L., Gui J.F (2017) Distinct and cooperative roles of *amh* and *dmrt1* in self-renewal and differentiation of male germ cells in zebrafish. *Genetics* **207**:1007-1022

27. **Liu C.K.** (1944) Rudimentary hermaphroditism in the symbranchoid eel Monopterus javanensis. *Sinensia* 1-8

28. **Liu J.K.,** Gu G.Y. (1950) Morphological changes in the gonad of *Monopterus* during sex reversal. *Science* **2**:91

29. **Liu L,** Guo M, Lv X, Wang Z, Yang J, Li Y, Yu F, Wen X, Feng L, Zhou T (2021) Role of Transient Receptor Potential Vanilloid 4 in Vascular Function. *Front. Mol. Biosci* **8**:677661 https://doi.org/10.3389/fmolb.2021.67766

30. **Liu X.Y.,** Wang L.C (1987) The relationship between sex and age, body length, and weight. *Freshwater fisheries* **6**:12-15

31. **Lu J.,** Huang S., Wei S., Cheng J., Li W., Fei Y., et al. (2025) Heat inducible nuclear translocation of Kdm6bb drives temperature dependent sex reversal in Nile tilapia. *PLoS Genet* **21**:e1011664 https://doi.org/10.1371/journal.pgen.1011664

32. **Luo T.T,** Zeng G.X., Xia C.C., Hu Z.X., Xue L.Z., Sun Y.H. (2026) Surrogate production of ricefield eel sperm by germ cell transplantation in adult zig-zag eel. *Water Biology and Security*

33. **Luo ZY,** et al. (2023) TRPV4 activation mediates Ca^{2+} influx and facilitates infectious spleen and kidney necrosis virus (ISKNV) replication. *Journal of Virology* **97**:1-14

34. **Martínez-Pacheco M.,** Díaz-Barba K., Perez-Molina R., Marmolejo-Valencia A., CollazoSaldana P., Escobar-Rodríguez M., Sanchez-Perez M., Meneses-Acosta A., MartínezRizo A., Sanchez-Pacheco A.U., *et al.* (2024) Gene expression dynamics during temperature-dependent sex determination in a sea turtle. *Dev. Biol* **514**:99-108

35. **Mei J.,** Gui J.F (2015) Genetic basis and biotechnological manipulation of sexual dimorphism and sex determination in fish. *Science China Life Sciences* **58**:124-136

36. **Mo X.Y.,** Pang P.Y., Wang Y.L., Jiang D.X., Zhang M.Y., Li Y., Wang P.Y., Geng Q.Z., Xie C., Du H.N., *et al.* (2022) Tyrosine phosphorylation tunes chemical and thermal sensitivity of TRPV2 ion channel. *eLife* **11**:e78301 https://doi.org/10.7554/eLife.78301

37. **Mundt N.,** Spehr M., Lishko P.V (2018) TRPV4 is the temperature-sensitive ion channel of human sperm. *eLife* **7**:e35853 https://doi.org/10.7554/eLife.35853

38. **Shao C.W.,** et al. (2014) Epigenetic modification and inheritance in sexual reversal of fish. *Genome Research* **24**:604-615

39. **Shi J.,** Sheng D.L., Guo J., Zhou F.Y., Wu S.F., Tang H.Y (2024) Identification of BiP as a temperature sensor mediating temperature-induced germline sex reversal in C. elegans. *The EMBO Journal* **43**:4020-4048

40. **Song Y.L.,** Luo H.R., Tao B.B., Chen J., Luo D.J., Chen K.X., Zhou W.Z., Hu W (2022) Research Progress on the genetic basis of natural sex reversal and varieties breeding in *Monopterus albus*. *Acta Hydrobiologica Sinica* **46**:1205-1214 https://doi.org/10.7541/2022.2020.238

41. **Steinberg X.,** Lespay-Rebolledo C., Brauchi S (2014) A structural view of ligand dependent activation in thermoTRP channels. *Front. Physiol* **5**:171-185

42. **Tao Y. X.,** Lin H. R., Van der kraak G., Peter R.E. (1993) Hormonal induction of precocious sex reversal in the ricefield eel, *Monopterus albus*. *Aquaculture* **118**:131-140

43. **Tian H.F.,** Hu Q.M., Li Z. (2021) A high-quality de novo genome assembly of one swamp eel (Monopterus albus) strain with PacBio and Hi-C sequencing data. *G3-Genes Genomes Genetics* **11**:jkaa032

44. **Todd E.V.,** Ortega-Recalde O., Liu H., Lamm M.S., Rutherford K.M., Cross H., Black M.A., Kardailsky O., Graves M., Hore T.A., *et al.* (2019) Stress, novel sex genes, and epigenetic reprogramming orchestrate socially controlled sex change. *Science Advances* **5**:eaaw7006

45. **Vrenken K.S.,** Jalink K., van Leeuwen F.N., Middelbeek J. (2016) Beyond ionconduction: channel-dependent and –independent roles of TRP channels during development and tissue homeostasis. Biochim. Biophys. Acta, Mol. *Cell Res* **1863**:1436-1446

46. **Wang X.,** Lai F.L., Xiong J., Zhu W., Yuan B.F., Cheng H.H., Zhou R.J (2020) DNA methylation modification is associated with gonadal differentiation in *Monopterus Albus*. *Cell Biosci* **10**:129 https://doi.org/10.1186/s13578-020-00490-4

47. **Wang X.,** Lai F.L., Shang D.T., Cheng Y.B., Lan T., Cheng H.H., Zhou R.J (2021) Cellular fate of intersex differentiation. *Cell Death and Disease* **12**:388-402

48. **Wang W.B.,** Zeng B.P., Luo Y.S., Han Q., Liu L.G (2008) The relationship between sex and body length and weight of ricefield eel in DongTing lake area. *Journal of Hunan Agricultural University (Natural Sciences)* **34**:469-473

49. **Wang and Zeng** (2006) Report on Monopterus albus from Hainan Infected with *Pallisentis* (Neosentis) *celatus* and Eustrongylides Species. *Sichuan Journal of Zoology* **125**:13-20

50. **Weber C.,** Zhou Y.J., Lee J.G., Looger L.L., Qian G.Y., Ge C.T., Capel B (2020) Temperature-dependent sex determination is mediated by pSTAT3 repression of *kdm6b*. *Science* **368**:3030-306

51. **Whiteley S.L.,** Weisbecker V., Georges A., Gauthier A.R., Whitehead D.L., Holleley C.E (2018) Developmental asynchrony and antagonism of sex determination pathways in a lizard with temperature-induced sex reversal. *Sci. Rep* **8**:14892

52. **Whiteley S.L.,** Holleley C.E., Wagner S., Blackburn J., Deveson I.W., Marshall Graves J.A., Georges A (2021) Two transcriptionally distinct pathways drive female development in a reptile with both genetic and temperature dependent sex determination. *PLoS Genet* **17**:e1009465

53. **Whiteley S.L.,** Wagner S., Holleley C.E., Deveson I.W., Marshall Graves J.A., Georges A (2022) Truncated jarid2 and kdm6b transcripts are associated with temperature-induced sex reversal during development in a dragon lizard. *Sci. Adv* **8**:eabk0275

54. **Wu P.F.,** Wang X.F., Ge C.T., Jin L., Ding Z.H., Liu F., Zhang J., Gao F., Du W.G (2024) pSTAT3 activation of Foxl2 initiates the female pathway underlying temperature-dependent sex determination. *PNAS* **121**:e2401752121 https://doi.org/10.1073/pnas.2401752121

55. **Xiong Y.,** Wang S., Gui J.F., Mei J (2020) Artificially induced sex-reversal leads to transition from genetic to temperature-dependent sex determination in fish species. *Science China Life Science* **63**:157-159

56. **Yao Z.L.,** Fang Q.F., Li J.Y., Zhou M., Du S.J., Chen H.J., Wang H., Jiang S.J., Wang X., Zhao Y., *et al.* (2023) Alternative splicing of histone demethylase *kdm6b* mediates temperature-induced sex reversal in the Nile tilapia. *Current Biology* **33**:5057-5070

57. **Yatsu R.,** Miyagawa S., Kohno S., Saito S., Lowers R.H., Ogino Y., Fukuta N., Katsu Y., Ohta Y., Tominaga M., *et al.* (2015) TRPV4 associates environmental temperature and sex determination in the American alligator. *Sci. Rep* **5**:18581-18591

58. **Ye Y.Z.,** Li J.M., Li W.M., Fu X.N., Zhang J.J., Yang S.L., Du W.G (2025) Transcriptional control of male-specific pathway in temperature-dependent sex determination. *Science Bulletin* https://doi.org/10.1016/j.scib.2025.07.025

59. **Yamamoto,** et al. (2024) Trpv4-mediated apoptosis of Leydig cells induced by high temperature regulates sperm development and motility in zebrafish. *Communications Biology* **7**:96 https://doi.org/10.1038/s42003-023-05740-y

60. **Zhang Y.M.,** Luo T.T., Sun Y.H (2025) A temperature-induced sex reversal mechanism in ricefield eels. *Water Biology and Security* https://doi.org/10.1016/j.watbs.2025.100360

61. **Zhou L.,** Gui J.F (2016) Jian-Kang Liu: a pioneer of sex determination studies in vertebrates. *Protein Cell* **7**:1-3

62. **Honeycutt,** et al. (2019) Warmer waters masculinize wild populations of a fish with temperature-dependent sex determination. *Scientific Reports* **9**:6527 https://doi.org/10.1038/s41598-019-42944-x

63. **Gutzke W.H.N.,** Crews D (1988) Embryonic temperature determines adult sexuality in a reptile. *Nature* **332**:832-834

Peer reviews

Reviewer #1 (Public review):

Summary:

This preprint investigates the molecular mechanism by which warm temperature induces female-to-male sex reversal in the ricefield eel (Monopterus albus), a protogynous hermaphroditic fish of significant aquacultural value in China. The study identifies Trpv4 - a temperature-sensitive Ca^{2+} channel - as a putative thermosensor linking environmental temperature to sex determination. The authors propose that Trpv4 causes Ca^{2+} influx, leading to activation of Stat3 (pStat3). pStat3 then transcriptionally upregulates the histone demethylase Kdm6b (aka Jmjd3), leading to increased dmrt1 gene expression and ovo-testes development. This work aims to bridge ecological cues with molecular and epigenetic regulators of sex change and has potential implications for sex control in aquaculture.

Strengths:

(1) This study proposes the first mechanistic pathway linking thermal cues to natural sex reversal in adult ricefield eel, extending the temperature-dependent sex determination paradigm beyond embryonic reptiles and saltwater fish

(2) The findings could have applications for aquaculture, where skewed sex ratios apparently limit breeding efficiency

Weaknesses:

Although the revised manuscript represents an improvement over the original version, substantial weaknesses remain.

Scientific Concerns

(1) Western blot normalization and exposure: The loading controls (GAPDH) in Fig. S3C appear overexposed, as do several Foxl2 blots. Because these signals are likely outside the linear range, I am not convinced that normalization is reliable. This raises concerns about the validity of the quantified results.

(2) Antibody validation and referencing (Line 776): The authors need to refer explicitly to figures demonstrating antibody validation. At present, these data are provided only as a supplementary file that is not cited in the manuscript. In addition, the Sox9a antibody appears to yield indistinguishable signals in control and RNAi conditions, suggesting that it may not recognize eel Sox9a. This issue is not addressed by the authors. Furthermore, antibody validation Western blots should be quantified.

(3) Unclear sample sizes (N values): Sample sizes remain unclear for several figures:

(a) Fig. 3F - No N value is provided. Each graph shows three data points; does this indicate that only three samples were quantified? If ten samples were collected, why were all not quantified?

(b) Fig. 4 - No N values are reported.

(c) Fig. 5A - Again, only three data points are shown per group, despite the apparent availability of twelve samples. The rationale for this discrepancy is not explained.

(4) qRT-PCR normalization: The manuscript does not specify the reference gene(s) used for qRT-PCR normalization. Although expression levels are reported as "relative," neither the identity of the reference gene(s) nor the justification for their selection is provided.

(5) Specificity of key antibodies: While the authors have made some effort to validate anti-Amh, anti-Sox9, and anti-Dmrt antibodies, the results remain incomplete. The Amh and Dmrt antibodies detect reduced protein levels following knockdown of their respective targets, which is encouraging. However, the Sox9a antibody shows no difference between control and RNAi conditions, suggesting it does not recognize eel Sox9. This is not acknowledged in the manuscript. In addition, no validation data are presented for Foxl2. Antibody validation data must be clearly referenced in the main text and presented in an interpretable and quantitative manner.

(6) Immunofluorescence data quality: The immunofluorescence images remain difficult to interpret. I strongly encourage the authors to enlarge the image panels and to present monochrome images (white signal on black background). The current presentation severely limits interpretability.

(7) Unreferenced supplementary figure: Fig. S4 is included in the submission but is not referenced anywhere in the manuscript text.

(8) Fig. 5B image resolution: The micrographs in Fig. 5B are too small to allow meaningful evaluation of the data.

(9) Unexplained data inclusion (Fig. 5E): Fig. 5E includes a pERK blot that is not mentioned in the Results section. The rationale for including these data is unclear.

(10) Poor blot quality (Fig. S3C): The blots in Fig. S3C exhibit high background and overexposure. I am concerned about the reliability of the quantification shown in panel D.

(11) Poor blot quality (Fig. S5G): The Stat3 blots in Fig. S5G contain numerous white artifacts, raising concerns about their suitability for normalization in panel H.

(12) Missing controls (Fig. 6E): Fig. 6E lacks controls for HO-3867 and Colivelin treatments alone. Without these controls, it is not possible to determine whether the reported effects are meaningful.

(13) Graphical presentation: The use of a light blue-to-pink gradient in bar graphs throughout the manuscript does not aid interpretation. I recommend using more distinct colors (e.g., red, orange, green, blue, purple, gray, black) to improve clarity. In summary, the interpretation of the study remains limited by persistent issues related to data presentation, image quality, and reagent specificity.

https://doi.org/10.7554/eLife.108272.2.sa1

Author response:

The following is the authors’ response to the original reviews.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This study investigates the molecular mechanism by which warm temperature induces female-to-male sex reversal in the ricefield eel (Monopterus albus), a protogynous*

> *hermaphroditic fish of significant aquacultural value in China. The study identifies Trpv4 - a temperature-sensitive Ca^{2+} channel - as a putative thermosensor linking environmental temperature to sex determination. The authors propose that Trpv4 causes Ca^{2+} influx, leading to activation of Stat3 (pStat3).pStat3 then transcriptionally upregulates the histone demethylase Kdm6b (aka Jmjd3), leading to increased dmrt1 gene expression and ovo-testes development. This work aims to bridge ecological cues with molecular and epigenetic regulators of sex change and has potential implications for sex control in aquaculture.*
>
> *Strengths:*
>
> *(1) This study proposes the first mechanistic pathway linking thermal cues to natural sex reversal in adult ricefield eel, extending the temperature-dependent sex determination paradigm beyond embryonic reptiles and saltwater fish.*
>
> *(2) The findings could have applications for aquaculture, where skewed sex ratios apparently limit breeding efficiency.*

We thank you for the encouraging comments of our work, and answering your questions has greatly improved the quality of the manuscript.

> *Weaknesses:*
>
> *(A) Scientific Concerns:*
>
> *(1) There is insufficient replication and data transparency. First, the qPCR data are presented as bar graphs without individual data points, making it impossible to assess variability or replication. Please show all individual data points and clarify n (sample size) per group. Second, the Western blotting is only shown as single replicates. If repeated 2-3 times as stated, quantification and normalization (e.g., pStat3/Stat3, GAPDH loading control) are essential. The full, uncropped blots should be included in the supplementary data.*

We thank you for the critical comments. Now we have remade the bar graphs with individual data points, and added the sample size per group if possible. Quantification and/or normalization of the WB data based on at least two replicates were included. The representative uncropped blots have also been loaded as the supplementary data.

> *(2) The biological significance of the results is not clear. Many reported fold changes (e.g., kdm6b modulation by Stat3 inhibition, sox9a in S3A) are modest (<2-fold), raising concerns about biological relevance. Can the authors define thresholds of functional relevance or confirm phenotypic outcomes in these animals?*

We thank you for the inspiring comments. Most of the experiments were transient in nature, for instance, warm temperature treatment of fish for 3-4 days, the fold change of gene expression were modest.

We admit that there are some shortcomings in this work. The major one is lacking of data showing that Trpv4 inhibition/activation,or pStat3 inhibition/activation can cause a gonadal phenotype change, for instance, from ovary to ovotestis or causing females to intersex fish. We only showed that pharmacological or RNAi can lead to change in sex-biased gene expression or affect temperature-induced gene expression, but not gonadal transformation.

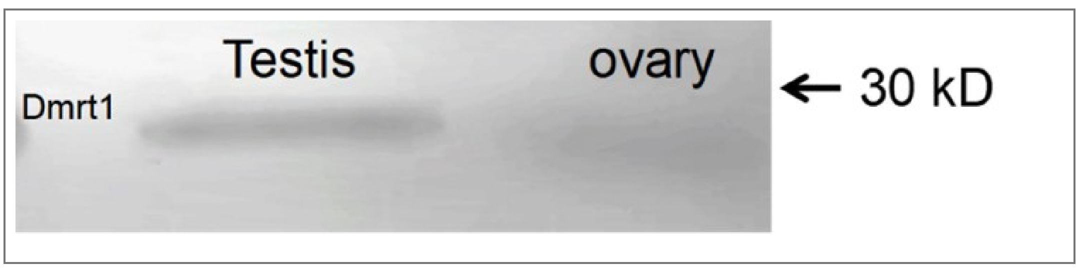
In natural population, the sex change of ricefield eel may take several months to one year or even longer. We propose that the magnitude and duration of temperature exposure promote sex change of ricefield eel by driving the accumulation of testicular differentiation genes in sufficient quantities. In experimental condition, to realize the gonadal phenotype change, animals may need to be under repeated pharmaceutical treatment (3 day interval treatment) for longer time to reach a threshold. However, long term treatment significantly increases the death rate of the animals, caused by stress or frequent manipulation.

Inspired by your comment, we are optimizing the experimental conditions in order to cause some phenotypic outcomes, thanks.

> *(3) The specificity of key antibodies is not validated. Key antibodies (Stat3, pStat3, Foxl2, Amh) were raised against mammalian proteins. Their specificity for ricefield eel proteins is unverified. Validation should include siRNA-mediated knockdown with immunoblot quantification with 3 replicates. Homemade antibodies (Sox9a, Dmrt1) also require rigorous validation.*

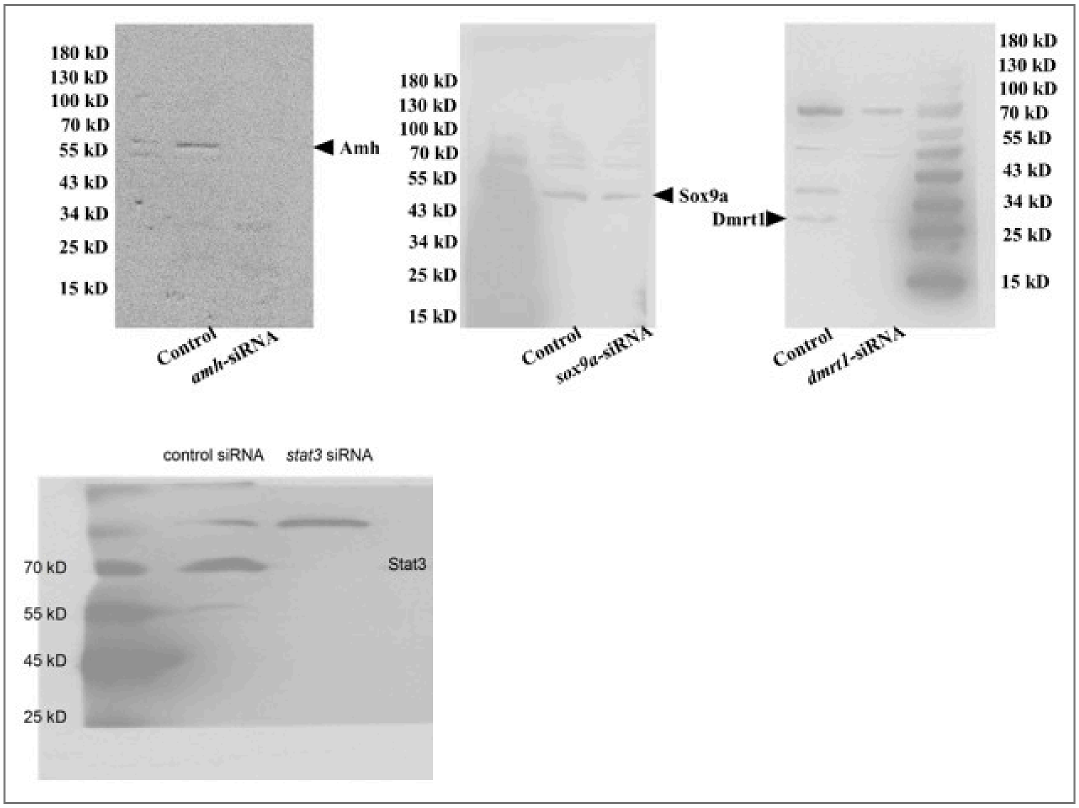
We thank you for the comments about the specificity of the antibodies. First,when choosing the commercial antibodies, we have compared the immunogen of the animal with the corresponding amino acids of ricefield eel, making sure that it was conserved to some extent (at least> 85% similarity). Second, we have referred the published work, where the antibodies have been proven to work in zebrafish, frogs, and turtles et al. This was true for pStat3 and Stat3 antibodies (Weber et al. 2020; Ge et al., 2024). Third, the specificity for each antibody was assessed using WB, based on the predicted size of the protein and the correct control setting.

For instance, we are very confident for the specificity for Dmrt1 antibody. First, Dmrt1 protein was readily detected in testes of males but barely detected in ovaries of females (Author response image 1). Second, Dmrt1 protein was not detected in ovary of fish at cool temperature, but clearly detected in nuclei of follicles in warm temperature-treated fish (Figure 3C, 4B), in line with our qPCR results. Third, by performing IF, Dmrt1 was not detected in females reared at lower temperature. By contrast, after warm temperature treatment or Trpv4 activation, it was detected in the nuclei in specific cell types but not everywhere (Figure 3E, 6C).



Author response image 1.

Although we have carefully evaluated the antibodies before experiments as described above, in response to your concerns, we went on to validate Amh, Dmrt1, Sox9a, and Stat3 antibodies using the corresponding siRNAs (Author response image 2). The results indicated that the antibodies, although not perfect, can be used in this work, as the expected band was gone or reduced in intensity. The experiments were repeated two times, and shown were representative.



Author response image 2.

> *(4) Most of the imaging data (immunofluorescence) is inconclusive. Immunofluorescence panels are small and lack monochrome channels, which severely limits interpretability. Larger, better-contrasted images (showing the merge and the monochrome of important channels) and quantification would enhance the clarity of these findings.*

We apologize for the poor quality of the IF images. At your suggestion, we have repeated the majority of the IF experiments, and imaging data with better quality were presented in the revised manuscript. Quantification of WB and IF was also included to enhance the clarity. Please see the revised manuscript, Thanks.

> *(B) Other comments about the science:*
>
> *(1) In S3A, sox9a expression is not dose-responsive to Trpv4 modulation, weakening the causal inference.*

We have repeated the experiments, and new data was included for the replacement of the old one in the revised manuscript.

> *(2) An antibody against Kdm6b (if available) should be used to confirm protein-level changes.*

We thank you for the nice suggestion. Unfortunately, current commercial antibody for Kdm6b is for mammals, which was not working in ricefield eel. At your suggestion, we are going to make one in future.

> *In sum, the interpretations are limited by the above concerns regarding data presentation and reagent specificity.*
>
> ***Reviewer #2 (Public review):***

> *Summary:*
>
> *This study presents valuable findings on the molecular mechanisms driving the female-to-male transformation in the ricefield eel (Monopterus albus) during aging. The authors explore the role of temperature-activated TRPV4 signaling in promoting testicular differentiation, proposing a TRPV4-Ca^{2+}-pSTAT3-Kdm6b axis that facilitates this gonadal shift.*

We thank you for the encouraging comments. Answering your questions has greatly improved our understanding of Trpv4 function in ricefield eel, and the quality of the manuscript.

> *Strengths:*
>
> *The manuscript describes an interesting mechanism potentially underlying sex differentiation in M. albus.*
>
> *Weaknesses:*
>
> *The current data are insufficient to fully support the central claims, and the study would benefit from more rigorous experimental approaches.*
>
> *(1) Overstated Title and Claims:*
>
> *The title "TRPV4 mediates temperature-induced sex change" overstates the evidence. No histological confirmation of gonadal transformation (e.g., formation of testicular structures) is presented. Conclusions are based solely on molecular markers such as dmrt1 and sox9a, which, although suggestive, are not definitive indicators of functional sex reversal.*

We thank you for pointing out this. The title has been changed to “Trpv4 links environmental temperature to testicular differentiation in hermaphroditic ricefield eel.”

> *(2) Temperature vs Growth Rate Confounding (Figure 1E):*
>
> *The conclusion that warm temperature directly induces gonadal transformation is confounded by potential growth rate effects. The authors state that body size was "comparable" between 25C and 33C groups, but fail to provide supporting data. In ectotherms, growth is intrinsically temperature-dependent. Given the known correlation between size and sex change in M. albus, growth rate-rather than temperature per se-may underlie the observed sex ratio shifts. Controlled growth-matched comparisons or inclusion of growth rate metrics are needed.*

We thank you for the critical comments. We have repeated the experiments, and have carefully compared the body length and weight, and results showed that there is no big difference between 25 and 33 degree groups. Please see Figure S1D-E, and the text in the last paragraph of “Warm temperature promotes gonadal transformation” section in the Results part.

> *(3) TRPV4 as a Thermosensor-Insufficient Evidence:*
>
> *The characterisation of TRPV4 as a direct thermosensor lacks biophysical validation. The observed transcriptional upregulation of Trpv4 under heat (Figure 2) reflects downstream responses rather than primary sensor function. Functional thermosensors, including TRPV4, respond to heat via immediate ion channel activity-typically measurable within seconds-not mRNA expression over hours. No patch-clamp or electrophysiological data are provided to confirm TRPV4 activation thresholds in eel gonadal cells.*

We thank you for the critical comments. The patch-clamp or electrophysiological experiments require special equipment and well-trained expert, unfortunately, our lab members and nearby collaborators have no experience in performing the kind of experiments. The Trpv4 is a well-known cation channel protein that is activated by moderate heat (> 27 degree). And a body of published work has demonstrated its role in the regulation of Ca^{2+} signals via change its configuration in response to temperature (J Physiol. 2017 Oct 25;595(22):6869–6885. doi: 10.1113/JP275052; Cell Death Dis 11, 1009 (2020). *https://doi.org/10.1038/s41419-020-03181-7*; Cell Death Dis 10, 497 (2019). *https://doi.org/10.1038/s41419-019-1708-9*; Cell calcium, *https://doi.org/10.1016/j.ceca.2026.103108*).

Consistently, warm temperature increased calcium influx within an hour, similar to the Trpv4 agonist treatment (Figure 2E, 5D), and addition of ion channel Trpv4 inhibitor prevents the calcium signals by war temperature treatment. Moreover, calcium signaling activity is closely linked with pStat3 activity and expression of sex-biased genes (Figures 5G, 6F). Although we did not show biophysical data, these results implied that Trpv4 is a thermosensor, and regulate the downstream pathway via the regulation of calcium signals, in line with it functions as an ion channel.

> *Additionally, the Ca^{2+} imaging assay (Figure 2F) lacks essential details: the timing of GSK1016790A/RN1734 administration relative to imaging is unclear, making it difficult to distinguish direct channel activity from indirect transcriptional effects.*

We have added more information for Ca^{2+} imaging assay (now Figure 2E and the corresponding text in Figure 2 legend, in the revised manuscript). In particular, we added the timing of treatment to better show that it was a direct effect.

> *(4) Cellular Context of TRPV4 Activity Is Unclear:*
>
> *In situ hybridisation suggests TRPV4 expression shifts from interstitial to somatic domains under heat (Figures. 2H, S2C), implying potential cell-type-specific roles. However, the study does not clarify: (i) whether TRPV4 plays the same role across these cell types, (ii) why somatic cells show stronger signal amplification, or (iii) the cellular composition of explants used in in vitro assays. Without this resolution, conclusions from pharmacological manipulation (e.g., GSK1016790A effects) cannot be definitively linked to specific cell populations.*

We thank you for the inspiring comments. We have performed IF experiments using Trpv4 specific antibodies (antibody specificity was confirmed). It was clearly shown that Trpv4 was expressed in a portion of follicle cells. To explore the identity of Trpv4-expressing somatic cells, we have performed double IF experiments using Trpv4 and Foxl2, a granulosa cell marker. The results (Figure 2H) clearly showed that Trpv4-expressing cells are a portion of Foxl2-positive granulosa cells. We propose that Trpv4-expressing granulosa cells may play an important role in sensing the temperature, and that Trpv4-expressing granulosa cells transdifferentiate into Sertoli cells by warm temperature exposure, because Dmrt1, a Sertoli cell marker, started within follicles in a typical granulosa cell location. Unfortunately, current Dmrt1/Trpv4 antibodies are both produced from rabbit. To overcome this, we are ordering mouse Dmrt1 antibodies, and in future we will perform Trpv4/Dmrt1 double IF to show if Dmrt1 positive cells co-localize with Trpv4 expressing cells. We would like to update the results to you once the antibody was available.

As our animal experiments (Figure 2H) have clearly shown the identify of Trpv4 expressing somatic cells, we did not repeat the experiments using explants, to explore the cellular composition of explants used in in vitro assays.

> *(5) Rapid Trpv4 mRNA Elevation and Channel Function:*

> *The authors report a dramatic increase in Trpv4 mRNA within one day of heat exposure (Figures 4D, S2B). Given that TRPV4 is a membrane channel, not a transcription factor, its rapid transcriptional sensitivity to temperature raises mechanistic questions. This finding, while intriguing, seems more correlational than functional. A clearer explanation of how TRPV4 senses temperature at the molecular level is needed.*

We appreciate you for your inspiring comments. Actually, we are also wondering about how trpv4 mRNA was regulated by warm temperature. First of all, the up-regulation of trpv4 mRNA is true, as evidenced by multiple pieces of data using qPCR and ISH experiments. It appears that ovarian cells respond to the temperature changes by increasing calcium influx via Trpv4 ion channel,as well as by increasing trpv4 mRNA expression levels.

Then, how trpv4 mRNA is regulated by heat? It is well-known that gene expression can be regulated by subtle temperature change via some direct temperature sensing genes (Haltenhof et al., 2020). We hypothesized that trpv4 is a downstream target of these thermosensors, displaying a mechanism similar to mammals. Actually, we have performed some experiments, and the preliminary data were obtained, which support our hypothesis.

Because the mechanistic explanation study is undergoing and not published, we chose not to discuss it in detail in the revised manuscript. We wish to report it by the end of this year, and by then are pleased to update you with the progress.

> *(6) Inconclusive Evidence for the Ca^{2+}-pSTAT3-Kdm6b Axis: Although the authors propose a TRPV4-Ca^{2+}-pSTAT3-Kdm6b-dmrt1 pathway, intermediate steps remain poorly supported. For example, western blot data (Figures 3C, 4B) do not convincingly demonstrate significant pSTAT3 elevation at 34C. Higher-resolution and properly quantified blots are essential. The inferred signalling cascade is based largely on temporal correlation and pharmacological inhibition, which are insufficient to establish direct regulatory relationships.*

We thank you for the critical comments. In response to your concerns, we have repeated experiments, and better resolution WB data with proper quantification were included in the revised manuscript. In particular, we convincingly demonstrate that 34 degree caused significant pStat3 elevation.

To directly establish regulatory relationship of the members, at your suggestion, we provided some genetic and molecular biology data to support our conclusion in the revised manuscript. For instance, we have knockdown the *stat3* gene by using siRNAs, and as shown in Figure 6F, we further showed that pStat3 is functionally downstream of Trpv4. Moreover, ChIP and luciferase assays were performed to show that pStat3 directly binds and activate *kdm6b* (Figure 7B-C). We have also performed various pharmacological inhibition to further strength our conclusion (Figures 6B-E).

> *(7) Species-Specific STAT3-Kdm6b Regulation Is Unresolved:*
>
> *The proposed activation of Kdm6b by pSTAT3 contrasts with findings in the red-eared slider turtle (Trachemys scripta), where pSTAT3 represses Kdm6b. This divergence in regulatory direction between the two TSD species is surprising and demands further justification. Cross-species differences in binding motifs or epigenetic context should be explored. Additional evidence, such as luciferase reporter assays (using wild-type and mutant pSTAT3 binding motifs in the Kdm6b promoter) is needed to confirm direct activation.*

We thank you for the inspiring comments. At your suggestion, we have performed luciferase assay using kdm6b promotor that is intact or mutated. The results were in favor of our statement. Please see Figure 7C and the related text.

> *A rescue experiment-testing whether Kdm6b overexpression can compensate for pSTAT3 inhibition-would also greatly strengthen the model.*

We thank you for the nice suggestion. It is technically challenging to perform kdm6b overexpression or any Kdm6b gain of function experiments (we have tried to make lentivirus, however, it was not working). There is no Kdm6b-specific agonists.

Inspired by you, we are establishing constitutive kdm6b transgenic ricefield eel. Although it require at least a year to allow the fish to grow up for functional experiments, once it was established, we can directly answer some important questions.

> *(8) Immunofluorescence-Lack of Structural Markers:*
>
> *All immunofluorescence images should include structural markers to delineate gonadal boundaries. Furthermore, image descriptions in the figure legends and main text lack detail and should be significantly expanded for clarity.*

We thank you for the critical comments. At your comments, we have first performed IF using beta-catenin as structural marker. However, the results were not good for some unknown reasons. Then, we used Vimentin as a structural maker, as it can label all the cells in gonads. Foxl2 was used as granulosa cell marker. Dmrt1 was used as Sertoli cell marker.

Some essential description was added in the figure legend as requested. Please see detail in the revised manuscript.

> *(9) Pharmacological Reagents-Mechanisms and References:*
>
> *The manuscript lacks proper references and mechanistic descriptions for the pharmacological agents used (e.g., GSK1016790A, RN1734, Stattic). Established literature on their specificity and usage context should be cited to support their application and interpretation in this study.*

These pharmacological agents have been used by others (Ge et al., 2017; Liu et al., 2021; Weber et al., 2020; Wu et al.,2024), and they are properly cited in the manuscript.

> *(10) Efficiency of Experimental Interventions:*
>
> *The percentage of gonads exhibiting sex reversal following pharmacological or RNAi treatments should be reported in the Results. This is critical for evaluating the strength and reproducibility of the interventions.*

We thank you for the critical and important comments. Actually another reviewer has asked the same question. We admit that this was the big shortcoming of the work, as we did not provide data demonstrating that Trpv4 inhibition/activation, or pStat3 inhibition/activation can cause a gonadal phenotype change, for instance, from ovary to ovotestis or causing sex reversal of fish. We only showed that pharmacological or RNAi can lead to alteration of sex-biased gene expression or affect temperature induced gene expression.

In wild population, the entire sex change of ricefield eel may take months to one year or even longer. We propose that the magnitude and duration of temperature exposure promote sex change of ricefield eel by driving the accumulation of testicular differentiation genes in sufficient quantities. In experimental condition, to realize the gonadal phenotype change, animals may need to be under repeated pharmaceutical treatment (3 day interval treatment) for longer time to reach a threshold, however, long term treatment significantly increases the death rate of the animals, caused by stress or frequent manipulation. Actually, my students have tried the experiments, unfortunately, either the number of sex-versing animals were small or the experiments lacked of repeat. So no percentage of gonadal transformation after treatment can be provided at this time, but we have indicated the number of samples when performing molecular experiments (showing expression of sex-biased genes).

Inspired by your important comment, we are optimizing the experimental conditions in order to cause some phenotypic outcomes. By then, the percentage of gonads exhibiting sex reversal following pharmacological or RNAi treatments can be calculated, showing the biological significance.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Editorial Concerns:*
>
> *(1) The term "sex reversal" should be clearly defined upfront as female-to-male, and the developmental consequences (e.g., increase in body size post-transition) should be explicitly stated early in the introduction.*

We thank our editorial for pointing out this. We have added those in the introduction Part. It reads “The species begins life as a female and then develops into a male through an intersex stage, thus displaying a female-to-male sex reversal during aging. Females are small in size (< 25 cm), and during and after sex change, there is a gradual increase in body size (> 55 cm for the majority of males).”

Additional information was shown in the first and second paragraph in the results Part.

> *(2) The manuscript references skewed sex ratios in cultured ricefield eel but fails to specify the direction (e.g., too many males or females). This should be clarified to contextualize the biological and commercial problem.*

According to your suggestion, we now added additional information, and it reads “The reproductive mode of ricefield eel, which leads to much more females than males in spawning season, severely affects the sex ratio, and decreases the productivity of broodstock. Moreover, adult females lay limited eggs (~200) due to its small size.”

> *(3) Define TSD (temperature-dependent sex determination) upon first use, not at the second mention.*

We have checked this, and make sure it was properly done.

> *(4) The phrase "quality fries for aquaculture" should be reworded or defined; it is unclear to non-specialists.*

We thank you for pointing out this. Now it reads “adult females lay limited eggs (~200) due to its small size, which is a limiting factor for massive production of seedling for aquaculture industry”.

> *(5) Several in-text citations (e.g., Weber 2020, Wu 2024) are absent from the bibliography.]*

We have double checked the reference, thanks.

> *(6) The inclusion of page and line numbers would facilitate peer review.*

We have now shown the page and line.

> *(7) The discussion is written vaguely. Clarify species names when discussing comparative biology and consider breaking down complex sentences to aid comprehension for a broad audience, such as that of eLife.*

We have added the species name, and try our best to use concise expression. Thanks.

https://doi.org/10.7554/eLife.108272.2.sa0